EIGSEP Memo 008

Marjum Pass 2026-07: image-domain modeling of the sky, the ground and the Sun

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Living document. This version: 2026-10-20

Abstract

How calibrated Marjum spectra are modelled in the image domain: as the beam-weighted sum of a Galactic sky map, a terrain at one uniform temperature below the DEM horizon, a receiver offset, and the Sun. The memo covers the model and the degeneracies its geometry leaves, the night fit to the calibrated 07-17/18 data, and the Sun. Box-air records solar radio bursts at the times independent spectrographs recorded them, which identifies solar emission in the data. The quiet Sun is harder: the ridges hide it for hours, so it appears only at a handful of terrain crossings. The one calibrated crossing, 15:54 UTC on 07-17, shows an excess 3.6 times the concurrent Sun's prediction (1.4 scaled by the template amplitude), while the raw data at the other crossings allow a Sun of the expected size but do not detect it convincingly. Each measurement is shown against the raw data around it. Finally, regressions on the rotating antenna give a repeatable ground term, too frequency-dependent to be the ground's temperature alone. Over all of phase C, raw power at 62–82 MHz carries a slow additive daily term that follows neither the pointing nor any recorded temperature; that it is reflected or picked-up sky does not reproduce from one day to the next. With a gain free to change within the hour, as the model requires, a phase-C ground term follows at 110–205 MHz, with errors from injected mocks; at the frequencies where the fit collapses it is not determined, and the fast gain follows the RF switch's temperature but not the receiver's measured gain. Every number, figure and table is computed by `make_products.py` in this directory from the products under `derived/ground_sky/` (Appendix A). The kelvin scale is memo 004's, the beam is HFSS (memo 005 measures it), and pointing is memo 003's.

1 Summary

- **The model** is linear at each frequency: sky pixels, one ground temperature, one offset per receiver regime and, optionally, the Sun's flux density. The columns are integrated through an assumed beam on the real geometry (Section 2). A uniform shift of sky and ground against the offset is an exact null direction.
- **The DEM stops short of the southern horizon.** Extending it 4 km south raises the horizon by up to 5.47° at 87.5 m and changes the predicted antenna temperature by up to 47 K, with the sky and beam fixed (Section 2, Fig. 1). The extended DEM and its horizon are now `derived/dem/v0004` and `horizon_profiles_v0004`, and every fit and product here now uses them.
- **The ground temperature is not measured by the calibrated data.** They hold one pointing, one height and one regime at night (165 two-minute bins), where $T_{\rm gnd}$ and the offset are degenerate. Only their sum, the pedestal, is determined: 12 to 90 K at 110–170 MHz. A forecast on the full campaign geometry shows that height changes are what would break the degeneracy (Table 1).
- **The data vary about twice as much as GSM through HFSS predicts**: the template amplitude is $a \approx 2.1$ (1.3–3.1 over the band). The model also does not reach the noise: median χ^2 per bin is 120, and 234 on held-out blocks.
- **When the Sun is visible.** The ridges stand 39° high at 352 m to the east and 17° at 171 m to the west. The Sun clears the ridge at **15:54 UTC on 07-17** (87.5 m), 3.5 h after it reaches 0° altitude, and sets behind the ridge at **01:21 on 07-18** (91 m). The calibrated data therefore see the direct Sun only from 15:54 to 16:22 on 07-17. At these wavelengths the edge diffracts, so the Sun fades in over 10–30 min.
- **Box-air sees the Sun's radio bursts.** At the times e-CALLISTO spectrographs recorded type III bursts, raw box-air shows emission drifting from high to low frequency within seconds, bright at 45–85 MHz as those stations saw it. 5 of 12 bursts with data pass a test that 5.2% of random daytime times pass ($p = 2.2 \times 10^{-4}$). That identifies solar emission in our data physically (§6.5).

- **The quiet Sun across days.** All 6 terrain crossings of phase C are shown in their raw context; 3 of them, on three days and three heights, can be fitted. They allow a Sun term of 3–16 SFU at 121–219 MHz, the size of the concurrent Sun (RSTN at 245 MHz, steady at 13–17 SFU from 07-15 to 07-18). They do not detect it convincingly, since raw power drifts by as much for other reasons (§6.7). None shows a jump as large as the calibrated 07-17 one.
- **The calibrated 07-17 sunrise excess is larger than the Sun.** After 15:54 the calibrated spectrum rises by a smooth, additive 20–25 K at 148–199 MHz (Fig. 5). That is 3.6 times what concurrent RSTN flux predicts (1.4 times scaled by a), and 2.0–3.2 times the other crossings at 199 MHz. A local fit gives $S_\odot$ from 7 SFU at 109 MHz to 32 SFU at 199 MHz; the same fit at the 0° sunrise, with the Sun behind the ridge, gives −6 to 1 SFU. Part of the excess is probably not the Sun. Something that changes as the box comes into sunlight would follow the same geometry.
- **Impulsive events in daytime are local**, not solar: their rate is the same with the Sun behind the ridge as with it visible (§6.6).
- **Not yet possible**: separating the Sun from the lit ground by timing, or from the sky and ground by the beam's sweep during the raster (§6.8). Below 90 MHz the calibrated result is unusable.
- **The ground from the rotating antenna.** When the antenna moves, the ground fraction swings while the gain stays put, so the ground/GSM ratio of a gain-free regression gives the ground's brightness in GSM kelvin. Three 91 m windows agree on it. It falls from 490 K at 125 MHz and 336 K at 139 MHz to 195–207 K at 150–175 MHz and reaches thousands below 85 MHz, which a reflected-sky term does not explain (§7). A joint fit across frequency and the sky-calibrated parked nights do not separate the ground, and low-order corrections to GSM are not separable from ground-fixed structure.
- **The low-frequency daily term over phase C.** A model of all of phase C needs a slow additive term at 62–82 MHz beyond a constant ground. It does not follow the sky of the current pointing, and none of five housekeeping streams reproduces it. Diffuse reflection of the sky fits it over the whole phase, and sky shifted in sidereal time does not, but only while the ground's level is tied to the reflection coefficient. The real-sky preference and the coefficient ($R = 1.34$–1.52 on 07-15, -0.06 to 0.17 on 07-16) do not reproduce between days. So the sky origin is withdrawn. All of this holds with gain knots every 3 h. With knots every hour or faster the gain follows the same daily shape and the term is no longer needed, which a mock with a real term also shows. So whether it is an additive term at all is open (§7.5).
- **The ground term with a fast gain.** Over phase C the model needs a gain that changes within the hour: with 3-h knots $T_{\rm gnd}$ comes out 2.2–6.7 times lower, and an injected mock with the 20-min gain refitted with 3-h knots reproduces those values to within 109 K. With 20-min knots, $T_{\rm gnd}$ falls from 693 K to 185 K over 110–205 MHz, with injection errors of 26–323 K, 0.7–7.6 times the fit's own. At 51, 66, 74, 188, 195, 211, 219, 227 MHz the fit has two solutions the data cannot tell apart, one with the GSM amplitude collapsed, and no ground term is quoted; 59, 234 MHz are unstable. On 07-17 the fast gain falls as the RF switch warms (−2.8%/K); on 07-15/16 the sign is the same but the relation rests mostly on the warmest knots. No other recorded stream follows it. Where the receiver's gain after the RF switch is measured (07-17, 87.5 m) it moves by about 1.1% and does not follow the sky, while the fitted gain moves by up to 27.7% and does; so the fast gain is a gain change before the switch or the fit absorbing model mismatch. A fast gain shared across frequencies, fitted jointly to convergence, removes most of the collapse and gives nearly the same high-band ground term whether or not the low band shares it (§7.6).
- **A solar burst at the 07-17 sunset.** Box-air alone shows a broadband excess below 100 MHz during a C1.5 flare (23:51–00:09 UTC). It ends while the Sun's disk goes behind the ridge, before the antenna's elevation steps (§6.7).

2 The model

At each frequency, every time bin t is modelled as

$$T(t) = \sum_p A_{\rm sky}[t,p]\,T_{\rm sky}[p] + A_{\rm gnd}[t]\,T_{\rm gnd} + T_{\rm off}[r(t)] + A_\odot[t]\,S_\odot, \tag{1}$$

with the sky in Galactic HEALPix pixels ($N_{\rm side} = 8$), the ground everything below the DEM horizon at one temperature, an offset per receiver regime r, and the Sun's flux density $S_\odot$ in SFU ($10^{-22}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$). The sky is written $T_{\rm sky} = a \cdot \mathrm{GSM} + \delta$. The amplitude a is free; it absorbs the front-end loss (calibrated temperatures are at reference plane P, memo 004) and any GSM scale error. Only δ has a prior, a Gaussian of 10% of GSM.

Integration. The sky and ground columns are integrated on a topocentric grid ($N_{\rm side} = 64$). For each distinct pointing and height era, the beam (HFSS total power here) is rotated by the mount, `mount_rotation`$(\mathrm{az} + \mathrm{az}_0, \mathrm{el} + \mathrm{el}_0, \psi)$, with $\mathrm{az}_0 = -16.04°$, $\mathrm{el}_0 = +0.93°$ and $\psi = 142.2°$ from the reference beam fit `empirical_raster_v0013` (memo 005), the frame in which its empirical beam was fitted. It is cut at that era's DEM horizon (`horizon_profiles_v0004`, traced from the antenna of geometry release `v0004`, memo 002, on the DEM mosaic `derived/dem/v0004`, which extends v0003 6 km south). Only the map from grid directions to Galactic pixels changes with time. Rows are normalized by the full-sphere beam integral, so sky and ground weights sum to one. That makes a uniform shift of sky and ground, against the offset, an exact null direction: breaking it needs a known offset or a known sky. The solver (`eigsep_sim.design_matrix.solve`) reports which columns move along a null direction and marks them unconstrained.

Bearings. The DEM, and so the horizon profiles, are in NAD83(2011)/UTM 12N. Grid north there is 1.52° from true north. The fit rotates the horizon to true bearings; the first trials of 2026-10-01 did not.

Terrain coverage. Earlier versions of this memo traced the horizon only out to the edge of the DEM mosaic `derived/dem/v0003`, which reaches $N = -1000$ m on its southern side. To see what that edge costs, we extended the mosaic 4 km south on the same grid, keeping every original pixel, the origin, the resolution and the vertical datum, and traced the horizon again from the profile antenna position $(E, N, U) = (1656.2, 2031.8, 1778.6)$ m (the 91 m era of geometry release `v0004`) in the DEM frame. The 30, 87.5 and 91 m height labels subtract 61, 3.5 and 0 m from U. Bearings are true bearings, rotated from grid north by 1.52° as above, in 0.1° steps with 720 sub-samples of the mask. Only the DEM changes between the two traces: sky, beam, site, times, pointings and frequencies are the same. The southern terrain raises the horizon by up to 3.45°, 5.47° and 5.51° at 30, 87.5 and 91 m. At the eight observed night pointings and four frequencies (51–246 MHz), the predicted antenna temperature changes by at most 16, 47 and 48 K (Fig. 1). The 87.5 m case is the selected night (07-17, 04:22–11:23 UTC); the 30 and 91 m cases are synthetic height controls evaluated at those same times, not observations from those eras. These are differences between two model predictions, not measured errors in any calibrated spectrum. A third trace with 8 km of added terrain differs from the 4 km one by at most 4.1×10^{-12} K, at the nominal position and six ±3 mm displacements. Edge flags alone do not test coverage: the original DEM raised none at 30 m and still changed. The tracing is also not shown to be accurate to sub-mK: the finest bearing refinement tested changes the prediction by up to 7.6 mK, so the numbers above are diagnostic and not tied to any convergence bound. This paragraph and Fig. 1 compare that original DEM with the extension; every fit and product in this memo now uses the extended DEM v0004 below.

The extended terrain is now a product. `derived/dem/v0004` is v0003 plus 6 km to the south, on the same grid and keeping every v0003 pixel, and `curation/horizon_profiles_v0004` is the horizon `build_horizon_profiles.py` traces on it. Over v0003 it rises by at most 3.45, 5.47 and 5.51° at 30, 87.5 and 91 m, on 209 bearings at 87.5 m, and is unchanged elsewhere. Terrain outside v0004 cannot change it: `coverage_certificate.py` follows every bearing beyond the DEM edge on the USGS 1 arc-second DEM, out to the distance (at most 42 km) beyond which terrain below 4200 m would sit under the skyline, and finds none that rises above it, from the profile position at all three heights or from the eight displaced positions of the profile's sensitivity block (`curation/horizon_profiles_v0004_coverage.json`). The coarse terrain is raised by 8.5 m, the most by which it falls below the skylines traced inside the DEM; the smallest margin is then 0.99°, on southern bearings, and 2.57° in the other directions. The check covers this antenna position only.

The Sun column. The Sun is a point source:

$$A_\odot(t) = \frac{\lambda^2}{8\pi k}\, G(\hat{s}(t))\, V(t) \quad \text{(K per SFU)}, \tag{2}$$

for one linear polarization and an unpolarized source. Here G is the beam gain toward the Sun, with the same full-sphere normalization as the rows, and V is its terrain visibility. V is not read off the fixed horizon profile.

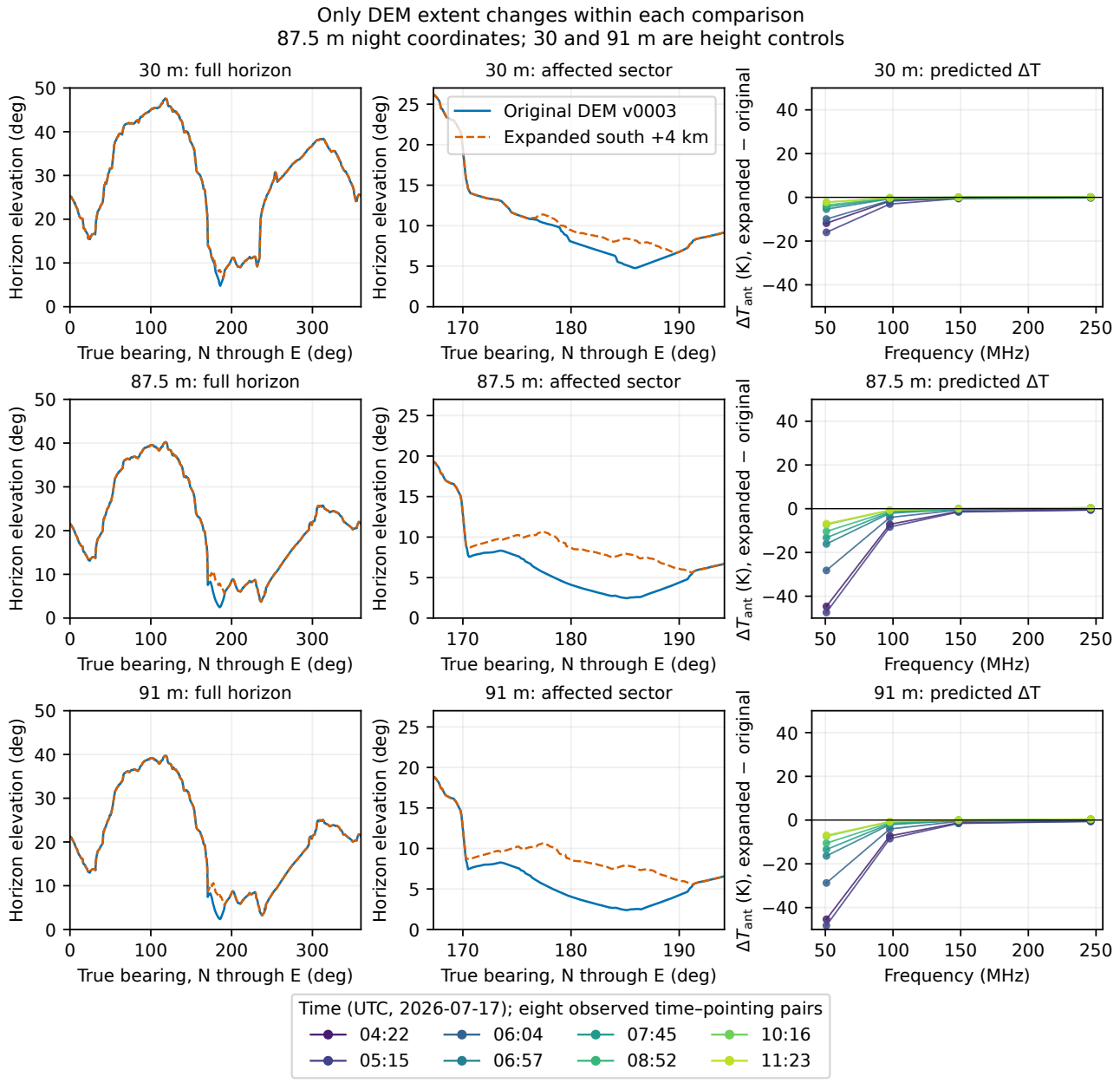


Figure 1: Effect of extending the DEM 4 km south, at the nominal antenna position, for the three height labels (rows). The horizon from the original DEM (`derived/dem/v0003`) and the extended DEM, with the affected southern sector enlarged, and the extended-minus-original predicted temperature at four frequencies, with the sky and beam fixed. Colours mark the eight observed night time/pointing pairs by UTC on 07-17; points are the evaluations and lines guide the eye. Within each pair only the DEM changes. The 30 and 91 m rows are synthetic height controls at the same times. The model azimuth is 0.865° for all eight pairs and the model elevations are 0.63–1.33° (`inputs/terrain_coverage/coordinates.csv`). Bearings are true; the horizon step is 0.1°.

The DEM is traced along the Sun's own bearing at each time, which gives the edge's elevation and its distance d. The ridges are 150–350 m away, so at these wavelengths the edge diffracts over $\sqrt{\lambda d/2} \approx 2$–$6°$, tens of minutes of solar motion. V is the Fresnel knife-edge power for the Sun's angle θ above the edge, $v = -\theta\sqrt{2d/\lambda}$: it is 0.25 at $\theta = 0$ and rises to 1 with lit-side ripples. The real ridge is rough and rounded, which smooths these ripples.

The knife-edge shape is an assumption. No measurement here tests it: its frequency-dependent onset, its value at the edge, its ripples. Every fit uses $A_\odot$ only as a template, with one fitted amplitude and, in the timing scans, one time shift. Figure 4 shows it against the data it was fitted to.

3 What the geometry can constrain

Before any data are used, `degeneracy_study.py` builds the design matrix on the campaign's real pointing and heights. It reports Fisher errors on $T_{\rm gnd}$ with radiometer noise in 120 s bins (Table 1).

- With the sky free, $T_{\rm gnd}$ is never constrained. That holds even with a known offset: within one pointing and era the ground column is constant and trades against a uniform sky shift.
- With a 10% GSM prior and the offset free, $T_{\rm gnd}$ and the offset are anticorrelated at $\rho = -0.95$ to -1.00. Height changes, which move the horizon, break the degeneracy: all of phase C would give 0.8–16 K.
- The calibrated windows alone are weak unless the offset is known.

The study also simulates and recovers. With HFSS as the true beam the recovered $T_{\rm gnd}$ is unbiased. With the empirical beam as truth and HFSS assumed, it is biased by up to 39 K (16σ) at 78 MHz, while χ^2 per row moves only from 3.1 to 3.5. A good χ^2 cannot vouch for $T_{\rm gnd}$.

slice	58.6	78.1	101.6	148.4	199.2
10% GSM sky prior, offset free					
parked, 30 m only	100.4	47.8	26.6	15.9	17.9
parked, all eras	21.4	10.0	5.4	2.8	1.1
all of phase C	16.0	7.4	3.9	1.7	0.8
tcal windows only	186.7	46.3	13.6	86.8	2.3
10% GSM sky prior, offset known to 1 K					
parked, 30 m only	7.9	3.7	2.2	2.0	1.7
parked, all eras	5.0	2.5	1.6	1.4	0.8
all of phase C	3.3	1.7	1.2	1.0	0.6
tcal windows only	11.4	5.4	2.9	2.2	1.2

Table 1: Forecast $\sigma(T_{\rm gnd})$ in K at five frequencies (MHz), from the degeneracy study (`summary.json` in `data-analysis`; radiometer noise, 120 s bins). "–": unconstrained. This forecast predates `tcal` v0003 and uses the v0000 calibrated windows.

4 Data

Box-air, antenna-switch rows calibrated by `tcal@v0003` (memo 004), which cover 07-17 04:11–16:22 (87.5 m, receiver regime rx-A) and 07-18 01:27–02:56 (91 m, rx-B). Samples flagged by `flags@v3` (memo 006) are dropped, on any bit but the advisory bit 9. From 07-17 15:36 the transmitter's comb channels (every eighth channel) are dropped whatever the flags say. At each HFSS beam frequency the remaining channels within ±2 are averaged. Rows are then averaged into static bins of one era, regime and pointing (az and el each within 1°): 302 two-minute bins (275 rx-A, 27 rx-B; product `binned_v0001`) and 615 one-minute bins (`binned_v0002`). Each bin's noise is the standard error of its rows. Frequencies whose in-bin scatter exceeds ten times the radiometer expectation are not fitted, which removes FM and leaves 44.

5 The night fit

`fit_v0013` fits Eq. 1 without the Sun, to the 165 bins with the Sun below $-10°$ (Fig. 2, Table 2). All of them are rx-A, at one pointing and one height.

- $T_{\rm gnd}$ **is not measured.** It is exactly degenerate with the offset, as Section 3 predicts. At 25 frequencies the $\pm 1°$ elevation wobble formally separates them. The values there run from -7053 to 11022 K: fitted model error, not a temperature. The pedestal $\langle f_{\rm gnd}\rangle T_{\rm gnd} + T_{\rm off}$ is determined: 12 to 90 K at 110–170 MHz.
- **The sky amplitude is** $a \approx 2.1$, measured to ± 0.02 at each frequency. Two explanations remain open: the `tcal` kelvin scale (it assumes the nameplate noise-source excess, memo 004), or a sky fraction that HFSS underestimates. The fit cannot tell them apart.
- **The model is not adequate at the noise level**: median χ^2 per bin is 120, and 234 on held-out alternating 30-min blocks.
- **The morning behind the ridge looks like night.** Adding the bins from 0° sunrise to the ridge crossing, `fit_v0014` (264 bins), raises the median χ^2 only from 120 to 136.

product	bins fitted	bins	freqs	median χ^2	median held-out χ^2
`fit_v0013`	night (Sun below $-10°$)	165 / 302	44	120	234
`fit_v0014`	Sun behind the ridge allowed	264 / 302	44	136	254
`fit_v0015`	night, 60 s bins	319 / 615	44	110	240
`fit_v0016`	Sun modelled, all bins	302 / 302	44	486	35520

Table 2: The fit products. Every fit evaluates the model on all bins; "bins fitted" are those used to solve. `fit_v0016` fits the Sun column on every bin, including those with the transmitter on; its held-out χ^2 reaches 3×10^6, and it is not used.

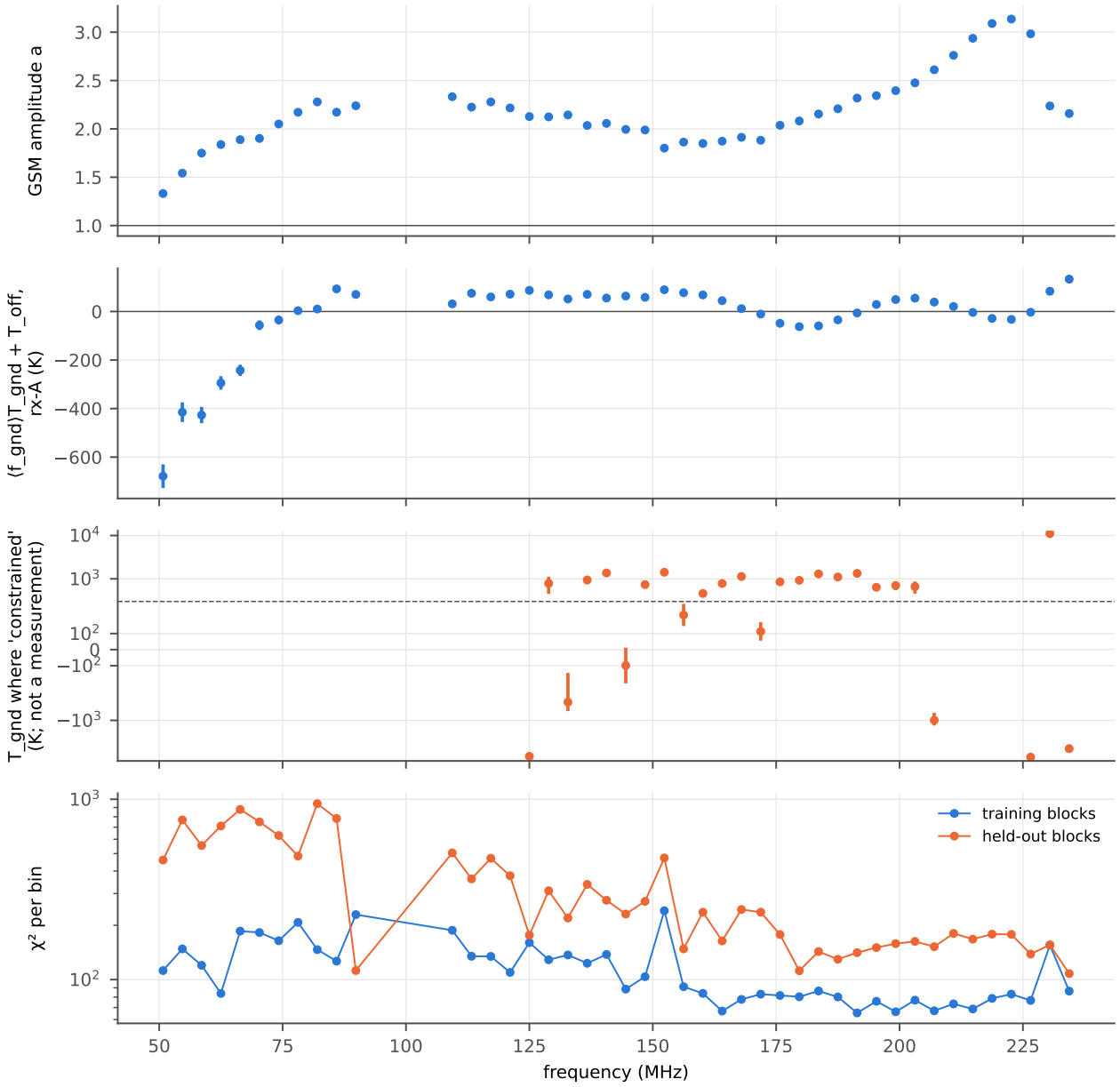


Figure 2: The night fit (`fit_v0013`). From the top: GSM amplitude; the rx-A pedestal; T_{gnd} where the solver calls it constrained (symlog axis; the dashed line is 300 K); χ^2 per bin on training and held-out blocks.

6 The Sun

6.1 When it is visible

Table 3 lists every time the Sun's centre crosses the DEM horizon or 0° altitude while each height era was flown in phase C. Figure 3 shows the horizons and the Sun's tracks. At 87.5 m the Sun clears the eastern ridge at 15:54 UTC on 07-17, at 39° altitude and azimuth 94°, with the edge 352 m away. That is 3.5 h after it reaches 0° altitude at 12:26. At 91 m it sets behind the western ridge at 01:21 on 07-18 (17°, edge 171 m), 1.5 h before 0° (02:53).

Two consequences follow. The rx-B window starts at 01:27, after the terrain sunset, so of the calibrated data only 07-17 15:54–16:22 sees the direct Sun. And the 2026-10-01 trials' observation that the Sun "lifts 199 MHz within an hour of rising" cannot be the direct Sun: that hour is behind the ridge.

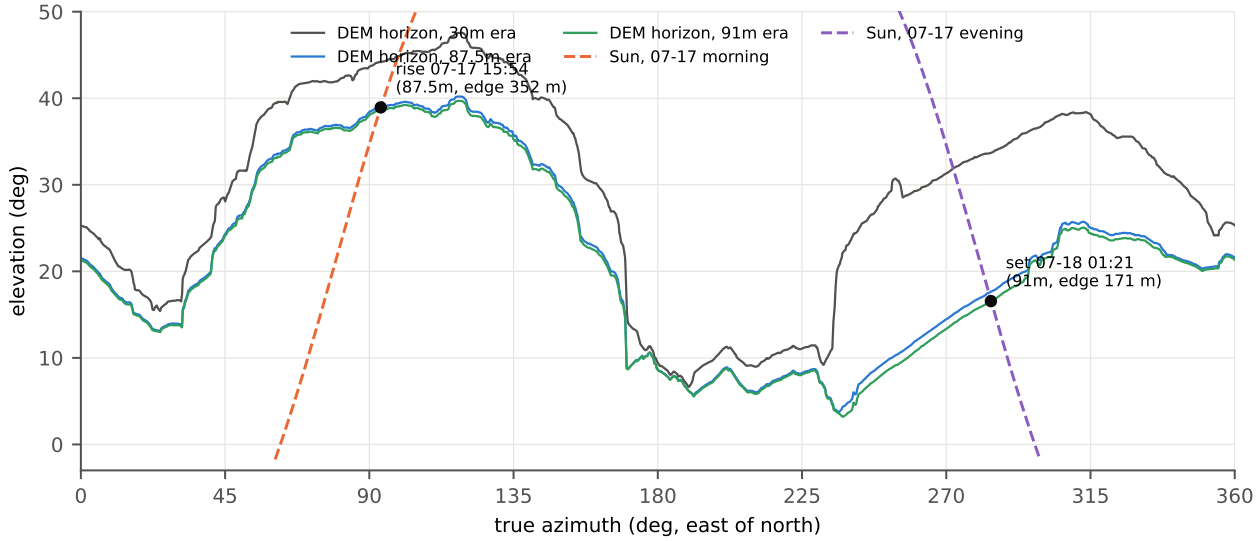


Figure 3: DEM horizon against true azimuth for the three heights (`horizon_profiles_v0004`, rotated from grid to true bearings), and the Sun's tracks on 07-17. Dots mark the terrain crossings in the calibrated eras.

era	event	horizon	UTC	Sun alt (deg)	Sun az (deg)	horizon (deg)	edge (m)
30m	set	flat	07-15 02:54:57	-0.0	298.3	0	–
30m	rise	flat	07-15 12:24:47	0.0	61.8	0	–
30m	rise	terrain	07-15 16:23:57	44.8	98.4	44.8	335
30m	set	terrain	07-16 00:01:47	32.0	272.6	32.0	151
30m	set	flat	07-16 02:54:17	-0.0	298.1	0	–
30m	rise	flat	07-16 12:25:27	0.0	62.0	0	–
30m	rise	terrain	07-16 16:24:17	44.8	98.7	44.8	335
30m	set	terrain	07-17 00:01:37	31.9	272.4	31.9	151
87.5m	set	flat	07-17 02:53:44	-0.0	297.9	0	–
87.5m	rise	flat	07-17 12:26:14	0.0	62.3	0	–
87.5m	rise	terrain	07-17 15:54:34	38.9	93.6	38.9	352
91m	set	terrain	07-18 01:21:30	16.6	283.9	16.6	171
91m	set	flat	07-18 02:53:10	-0.0	297.6	0	–

Table 3: Sunrises and sunsets during phase C, by height era (`sun_events_v0004`). "terrain": the Sun's centre crosses the DEM horizon traced along its bearing; "flat": 0° altitude. Geometric, with no refraction.

6.2 The test around 15:54

The test reads `fit_v0015`, a night fit on one-minute bins. Every bin with the Sun above 0° is out of sample there, so its residual (data minus fitted sky, ground and offset) is a prediction. That residual drifts by tens of kelvin over hours, as the night model is carried into the morning. So each frequency is fitted locally, from 14:30 to 16:23 UTC, as

$$\mathrm{resid}(t) = c_0 + c_1(t - t_{\mathrm{ridge}}) + S_\odot\, A_\odot(t - \tau).$$

The transmitter of the beam-mapping raster is on 15:36–16:14, with short gaps (box-gnd transitions, memo 001). Its bins are never fitted. That leaves 62 bins, of which only 7 (16:14–16:22) have the Sun above the ridge. Errors on $S_\odot$ are scaled by $\sqrt{\chi^2/\mathrm{dof}}$.

Each of the following checks could have come out the other way:

1. **A null at 0° sunrise.** The same fit around 12:26 (97 transmitter-off bins), with a Sun column that ignores the terrain, should give $S_\odot \approx 0$ if the ridge hides the Sun.
2. **Timing.** χ^2 against a shift τ of the Sun column should be smallest near $\tau = 0$.
3. **Spectrum.** $S_\odot(\nu)$ should be positive and smooth, of the order of the quiet-to-active Sun.

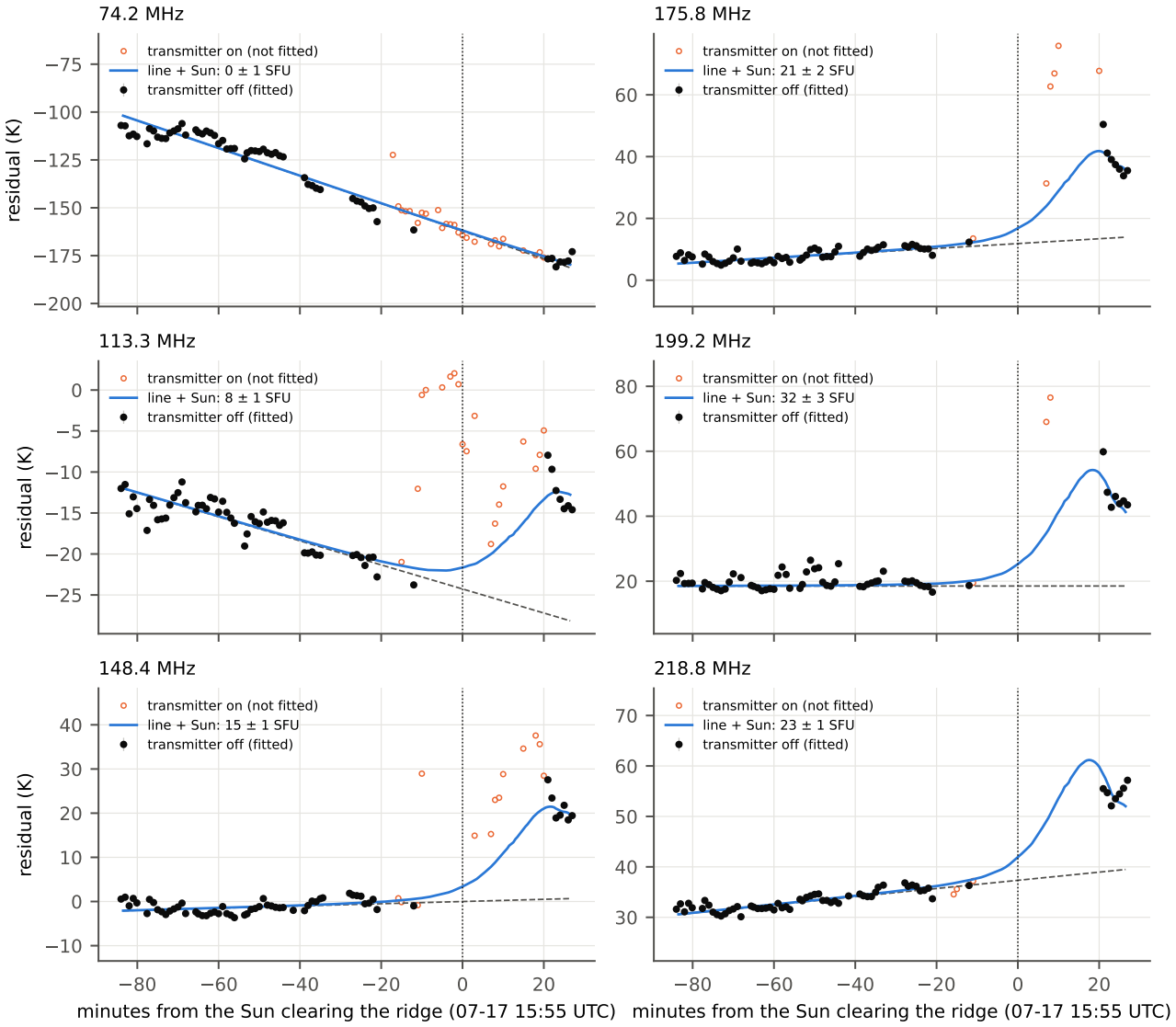


Figure 4: Out-of-sample residuals of the night fit around the ridge crossing (dotted line), at six frequencies. Filled: transmitter off, fitted. Open: transmitter on, not fitted. Dashed: the fitted line alone. Blue: line plus Sun.

6.3 Result

- **Spectrum.** At 31 frequencies between 105 and 235 MHz, $S_\odot$ runs from 7 to 33 SFU (31 positive), rising smoothly: 7±1 at 109 MHz, 15±1 at 148, 32±3 at 199 and 23±1 at 219 MHz (Fig. 6, Table 4).
- **Null.** At 0° sunrise the same fit gives −6 to 1 SFU over the same band. No step appears when the Sun rises behind the ridge.
- **Timing.** χ^2 is smallest at $\tau = -3$ min, and within 5 of that minimum from −7 to +3 min. A shift of 20 min earlier costs $\Delta\chi^2/(\chi^2/\mathrm{dof}) = 69$ on average.
- **Kelvin scale.** If the factor $a \approx 2$ lies in the kelvin scale or the beam, it scales $S_\odot$ too. $S_\odot/a$ is 3, 8 and 14 SFU at 109, 148 and 199 MHz.
- **Below 90 MHz** the result is not usable. The residual drifts by tens of kelvin across the window there, and the null fails (grey and orange points below 90 MHz in Fig. 6).

The spectra themselves. Figure 5 sets aside the local fit and compares spectra. It averages the calibrated spectrum (weighted means) over two windows of transmitter-off bins. "Before" is −40 to −15 min from the ridge crossing, with the Sun behind the ridge (15 bins). "After" is the 7 bins with the Sun above it. Their difference is shown raw, and with the night model's predicted sky and ground change between the windows subtracted. That change is at most 9 K above 109 MHz, but up to 51 K below 87 MHz, where the difference is dominated by drift the model does not capture. The difference is compared with a Sun predicted with nothing fitted to these data. The prediction takes RSTN's concurrent 245 MHz flux (§6.4), extrapolated down in frequency with the MWA quiet-Sun shape (scaled by 0.85), times the Sun column averaged over each window.

- Above 109 MHz the spectrum rises by a smooth 20 K at 148 MHz and 25±1 K at 199 MHz. The flux this implies, 15 and 34 SFU, agrees with the local fit.
- The prediction is 5 and 5 K. The measured jump is 1.9–5.6 times the prediction over 31 frequencies (weighted mean 3.6). Scaled by a, the prediction falls short by a factor of 2.1 below 200 MHz and 1.2 above 205 MHz. The excess is flatter in frequency than the quiet-Sun shape.
- **Not a gain change.** A change in receiver gain would scale the whole spectrum, so $\Delta T/T$ would be constant. Instead it runs from 1 to 16% (coefficient of variation 0.49), while ΔT itself varies less (0.27). The jump is additive, like a new source, not multiplicative.

MHz	$S_\odot$ (SFU)	χ^2/dof	a	$S_\odot/a$	null at 12:26
109.4	6.9 ± 0.7	119	2.25	3.1	-1.2 ± 0.5
125.0	13.3 ± 1.2	207	2.11	6.3	-1.0 ± 0.6
148.4	14.5 ± 0.9	125	1.91	7.6	-2.9 ± 1.1
175.8	20.9 ± 1.6	79	1.98	10.5	-2.9 ± 1.7
199.2	32.4 ± 2.5	128	2.36	13.7	-2.9 ± 1.4
218.8	23.5 ± 1.4	125	2.96	7.9	-1.6 ± 1.2
234.4	25.6 ± 1.8	39	2.20	11.7	1.1 ± 1.2

Table 4: $S_\odot$ at selected frequencies: the local fit's χ^2/dof, the GSM amplitude a of `fit_v0015`, $S_\odot/a$, and the flat-horizon null at 12:26.

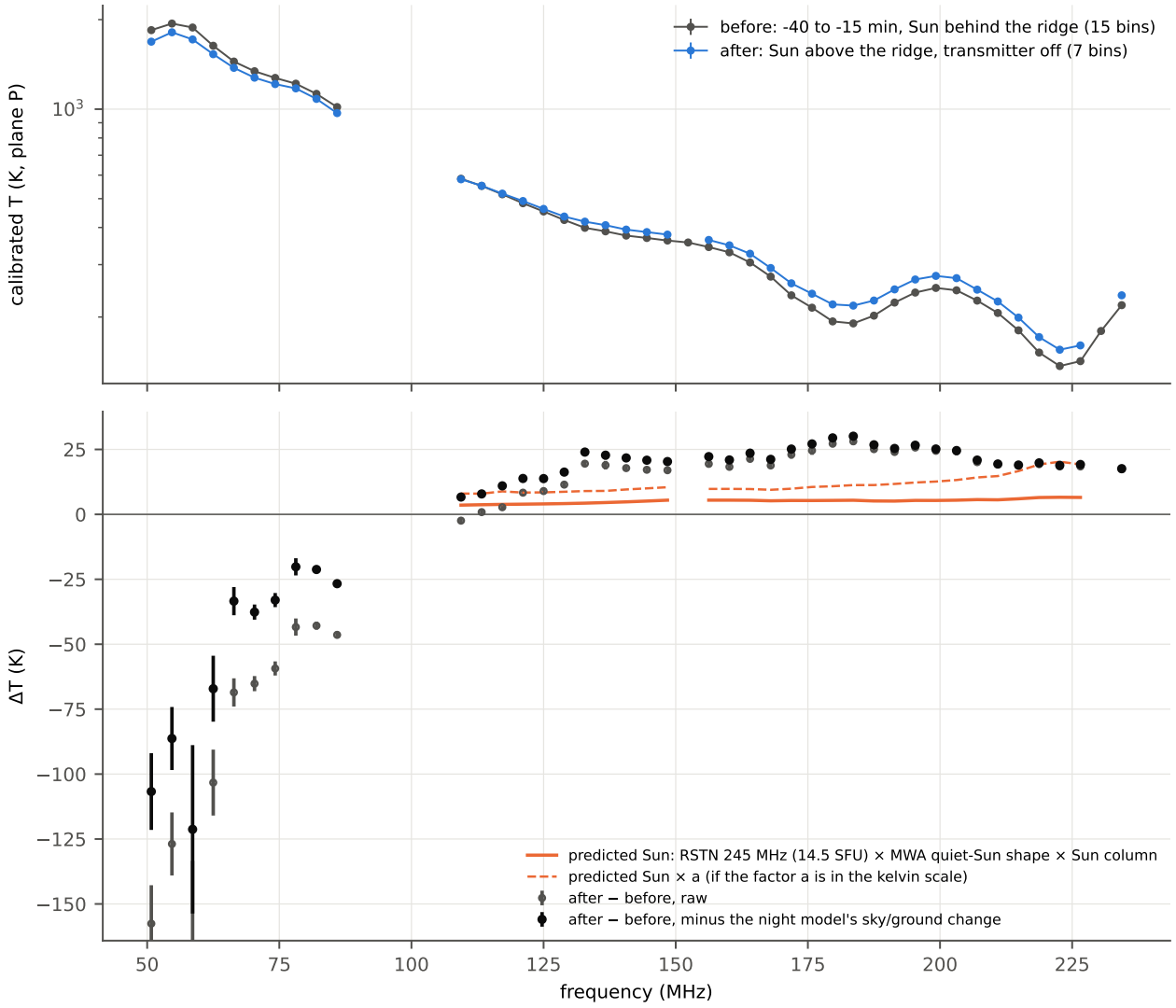


Figure 5: Top: calibrated spectra before the ridge crossing (Sun behind the ridge) and after it (Sun above, transmitter off). FM is blank. Bottom: their difference, raw (open) and with the night model's sky/ground change removed (filled). Orange: the Sun predicted from concurrent RSTN 245 MHz flux with the MWA quiet-Sun spectral shape, through the beam and the ridge; dashed: the same times the GSM amplitude a. Nothing in the orange curves is fitted to these data.

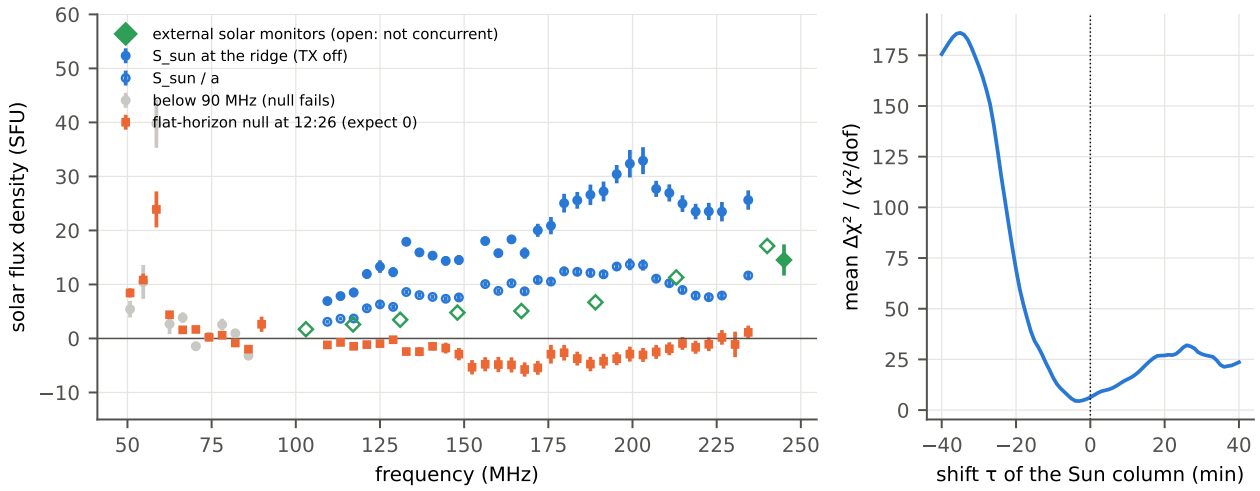


Figure 6: Left: $S_\odot$ from the transmitter-off fit at the ridge crossing (filled), divided by the GSM amplitude a (open), and the flat-horizon null at 12:26 (squares). Diamonds are external solar monitors (Section 6.4). Right: mean $\Delta\chi^2$ over frequencies against a shift of the Sun column; $\tau > 0$ is later than the DEM predicts.

6.4 Concurrent solar monitoring

The USAF Radio Solar Telescope Network (RSTN) measures the whole-Sun flux at 245, 410 and 610 MHz every second. Two stations recorded continuously through 15:54–16:22 UTC on 07-17. At 245 MHz Sagamore Hill reads 14.9 SFU (rms 0.8, 99th percentile 18) and San Vito 14.1 SFU (rms 0.7, 99th percentile 17), over 1651 and 1650 one-second samples. Palehua's reading is below 5 SFU for 52% of the window, as if not yet tracking the Sun; over the transmitter-off bins with the Sun up it reads 25 SFU. The stations' noon values for 07-17 at 245 MHz are Sagamore Hill 13, San Vito 13, Palehua 21, Learmonth 23 SFU. The stations disagree by more than the quoted 2–3 SFU; we take the two continuous stations' mean, 14.5 SFU, as the concurrent reference at 245 MHz, and note that Palehua reads higher.

The Sun was quiet: SWPC lists no flare or radio burst between 13:18 and 20:16 UTC, the 10.7 cm flux was 104 sfu (Penticton, 17:00 UTC), and e-CALLISTO spectrographs show no burst at 100–250 MHz in the window. Sagamore Hill's spectrograph recorded a series of type III bursts at 25–180 MHz from 12:14 to 13:18, inside the null window, when the Sun was behind the ridge. We have not looked for them in our data.

Nothing calibrated was found below 245 MHz for that day. For the spectrum's shape, the MWA's quiet-Sun measurement (Oberoi, Sharma & Rogers 2017, on 2013-09-03) gives 1.7, 3.5, 5.1, 11.3 SFU at 103, 131, 167, 213 MHz, rising with frequency as ours does, with 10–60% systematic error. At 234 MHz, the top of our band, $S_\odot$ is 26±2 SFU and $S_\odot/a$ is 12 SFU. RSTN gives 14.5 SFU at 245 MHz, a ratio of 1.8 ($S_\odot$) or 0.8 ($S_\odot/a$). Between 105 and 200 MHz, $S_\odot$ is 3.0–5.0 times the MWA quiet-Sun curve, and $S_\odot/a$ is 1.4–2.4 times it. The excess thus agrees with the concurrent Sun to within the unresolved factor $a \approx 2$. That agreement is not a test that separates the Sun from another effect of similar size at sunrise.

source	when	what it says
RSTN Sagamore Hill, 245 MHz	15:55–16:22	14.9 SFU mean, rms 0.8
RSTN San Vito, 245 MHz	15:55–16:22	14.1 SFU mean, rms 0.7
RSTN Palehua, 245 MHz	16:14–16:22	25.0 SFU mean; below 5 SFU (as if not tracking) for 52% of 15:55–16:22
RSTN noon values, 245 MHz	07-17 local noon	Sagamore Hill 13, San Vito 13, Palehua 21, Learmonth 23 SFU
Penticton F10.7	17:00	103.9 sfu observed
SWPC event list	all day	0007-0008 PAL RBR 245 820; 0150-0150 PAL RBR 245 260; 1214-1318 SAG RSP 025-180 VI/2; 2016-2053 SAG RSP 025-076 VI/2; 2025-2114 G19 XRA 1-8A B6.5 1.8E-03; 2030-2034 PAL RSP 025-075 III/2; 2214-2251 PAL RSP 025-055 VI/2; 2325-2325 PAL RSP 025-037 III/2
e-CALLISTO (6 stations)	15:45–16:30	no burst at 100–250 MHz (uncalibrated, visual)
MWA quiet Sun (Oberoi et al. 2017)	2013-09-03	103 MHz 1.71, 117 MHz 2.6, 131 MHz 3.46, 148 MHz 4.79, 167 MHz 5.08, 189 MHz 6.69, 213 MHz 11.3, 240 MHz 17.12 SFU

Table 5: External solar radio measurements for 2026-07-17 (`inputs/solar_reference.json`, with URLs).

6.5 Solar radio bursts in the raw data

A radio burst is a property of the Sun, recorded independently by other observatories. It is the most direct test of whether box-air receives solar emission. Type III bursts drift from high to low frequency within seconds and last tens of seconds; at 0.54 s per integration, box-air resolves them.

The reference list. `curation/solar_bursts_v0001.json` lists the 32 bursts that e-CALLISTO spectrographs recorded at two or more stations while the Sun was above the Marjum ridge (Table 3). It starts from e-CALLISTO's curated burst list for July 2026, a detector that requires three stations, with times to the minute. Spectrograms from 11 stations in daylight were then inspected to time each burst to about a second and to add bursts the list missed. All are type III or groups of type III. The stations cover 5–93 MHz, apart from one noisy one at 46–226 MHz, so the list cannot say whether a burst reached our higher frequencies.

Method. For each burst, `sun_bursts.py` reads raw box-air antenna rows from 90 s before to 90 s after the burst, with *no* RFI flags, since the flags mark positive excesses, which is what a burst is. It divides each channel by its median over that cutout. It then measures the excess in four bands during the burst, in robust standard deviations of the same band away from it, and finds when each 10 MHz sub-band peaks. Box-air antenna data exist at 12 of the 32 bursts; the rest fall in data gaps or calibration cycles (Table 6).

What the raw data show. Figure 7 shows three bursts as raw dynamic spectra, each channel divided by its median.

- **07-15 18:33 (30 m).** A narrow ridge of emission appears inside the e-CALLISTO interval and drifts from the 70 MHz range down to the bottom of our band. The 10 MHz sub-bands peak in sequence, at −8 MHz/s. In the light curves, 45–65 MHz rises by 99% of the raw power (115σ), while 170–235 MHz does not move (1σ).
- **07-15 18:23 (30 m).** Two components about 25 s apart, both low-frequency only, 119σ at 45–65 MHz. The brighter, later one drifts at −9 MHz/s.
- **07-17 20:33 (91 m, during the raster).** The brightest burst of the period (e-CALLISTO's "IIIG/3", a group). Several components sweep down to the bottom of our band. Sub-bands up to 120 MHz peak above 8σ and up to 220 MHz above 5σ, so the group reaches above the reference stations' bands. The raw power at 45–65 MHz rises by 120%. The raster's own modulation is visible behind it.

Test. A random time in daytime is often near one of the local spikes of §6.6, so a big excess alone is not a detection. The bursts e-CALLISTO saw are bright at 45–85 MHz and fainter above, while the local spikes are

broadband. The test therefore counts times that are bright at 45–65 MHz *and* quiet at 170–235 MHz ($z_{45–65} > 10$, $z_{170–235} < 5$), at the bursts and at 250 random times in the same Sun-up windows, at least 5 min from any listed burst (Fig. 8). 5.2% of the random times pass, against 5 of the 12 bursts with data; the chance of that many by accident is 2.2×10^{-4}. This test could have failed.

What this shows. Box-air records solar radio bursts. They come at the times independent stations recorded them, drift from high to low frequency as type III bursts do, and have the low-frequency-weighted spectrum those stations saw. That is a physical identification of solar emission in our data, not a coincidence in timing. It says nothing about the quiet Sun's flux, since the e-CALLISTO spectrograms are not calibrated.

burst start (UTC)	dur. (s)	e-C MHz	sta.	box-air excess (robust σ) 45–65	65–85	110–170	170–235	match
07-15 18:21:37	14	20–85	4	6	3	16	1	no
07-15 18:23:19	43	20–85	5	119	36	10	2	yes
07-15 18:33:02	19	10–85	6	115	4	1	1	yes
07-15 18:47:01	12	10–45	2	no box-air antenna data				–
07-15 22:45:10	5	10–45	2	no box-air antenna data				–
07-15 22:59:08	8	10–40	2	no box-air antenna data				–
07-15 23:49:00	10	15–50	2	11513	1796	3046	2184	no
07-16 16:23:55	19	20–85	6	19	1	2	4	yes
07-16 16:50:02	11	20–85	6	no box-air antenna data				–
07-16 16:54:58	13	20–90	7	61	11	72	19	no
07-16 17:16:06	26	20–85	6	no box-air antenna data				–
07-16 17:25:35	5	46–85	3	no box-air antenna data				–
07-16 17:26:22	10	20–85	6	no box-air antenna data				–
07-16 17:57:21	50	46–80	3	no box-air antenna data				–
07-16 18:33:13	31	20–85	6	no box-air antenna data				–
07-16 18:48:05	11	20–65	3	no box-air antenna data				–
07-16 19:17:07	16	20–85	5	no box-air antenna data				–
07-16 21:36:13	8	20–55	2	no box-air antenna data				–
07-16 22:05:44	10	15–45	2	no box-air antenna data				–
07-16 22:25:10	9	15–50	2	no box-air antenna data				–
07-16 22:28:10	16	15–60	3	no box-air antenna data				–
07-16 22:32:42	9	55–85	3	no box-air antenna data				–
07-16 23:58:59	34	20–75	2	17	9	16	20	no
07-17 20:16:26	14	30–85	4	44	9	3	0	yes
07-17 20:30:45	46	10–60	5	8	2	2	4	no
07-17 20:32:59	37	10–90	6	35	10	5	4	yes
07-17 20:34:14	23	10–85	5	5	2	0	1	no
07-17 21:21:22	99	10–60	4	4	4	2	2	no
07-17 22:26:42	49	15–45	2	no box-air antenna data				–
07-17 22:46:54	19	20–60	4	no box-air antenna data				–
07-17 22:51:02	21	15–40	2	no box-air antenna data				–
07-17 23:25:37	17	15–45	2	no box-air antenna data				–

Table 6: The 32 reference bursts (`curation/solar_bursts_v0001.json`): duration and frequency range at the e-CALLISTO stations, number of stations, and box-air's excess in four bands. "match": bright at 45–65 MHz and quiet at 170–235 MHz ($z_{45–65} > 10$, $z_{170–235} < 5$).

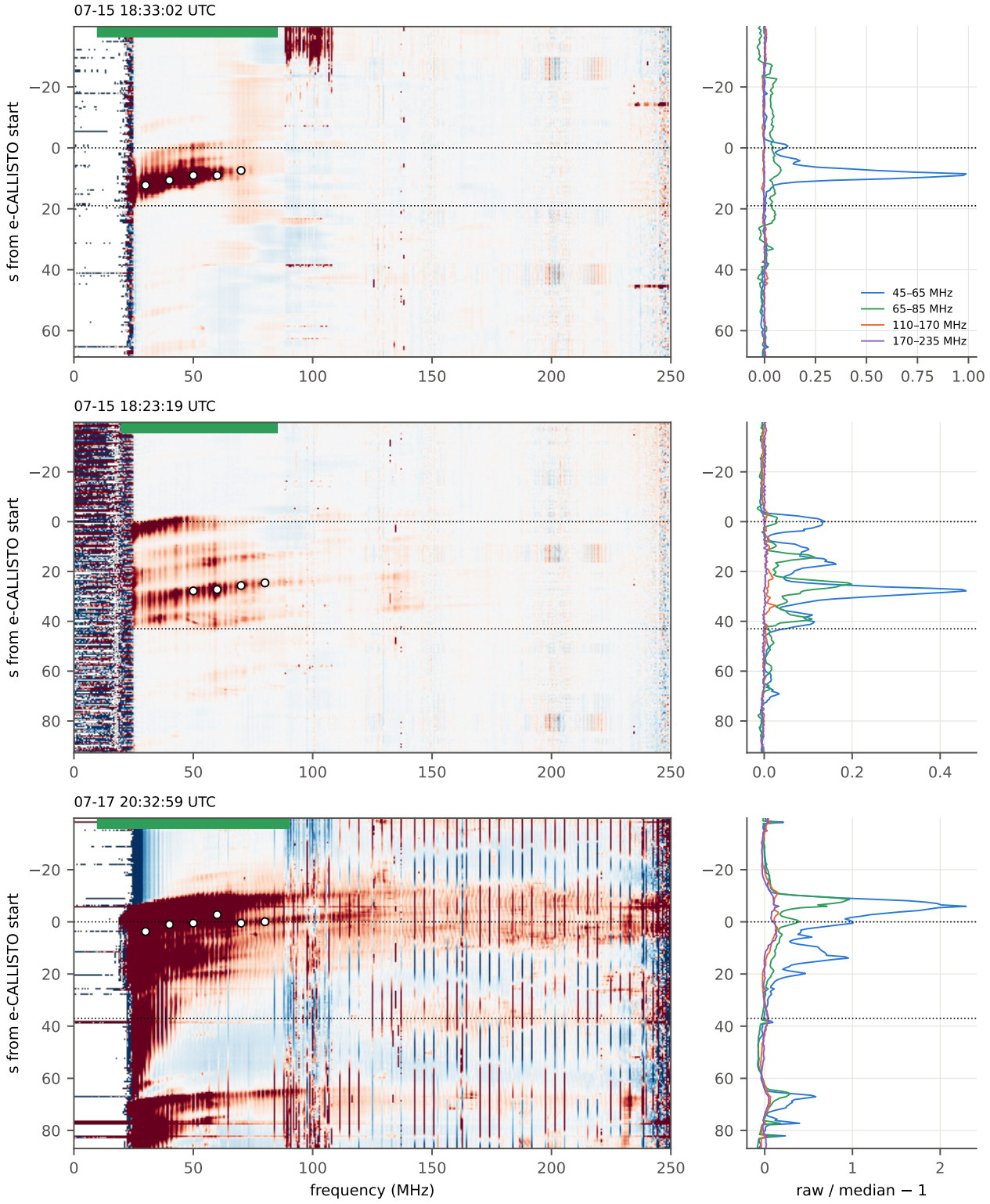


Figure 7: Three e-CALLISTO bursts in raw box-air data, time running down. Left: each channel divided by its median over the cutout, minus 1. Dotted lines are the burst's start and end at the reference stations, and the green bar along the top is their frequency range. Circles mark when each 10 MHz sub-band below 95 MHz peaks, where it is clear (above 8 robust σ) and within 8 s of the 45–65 MHz peak. Right: the same in four bands, on the same time axis.

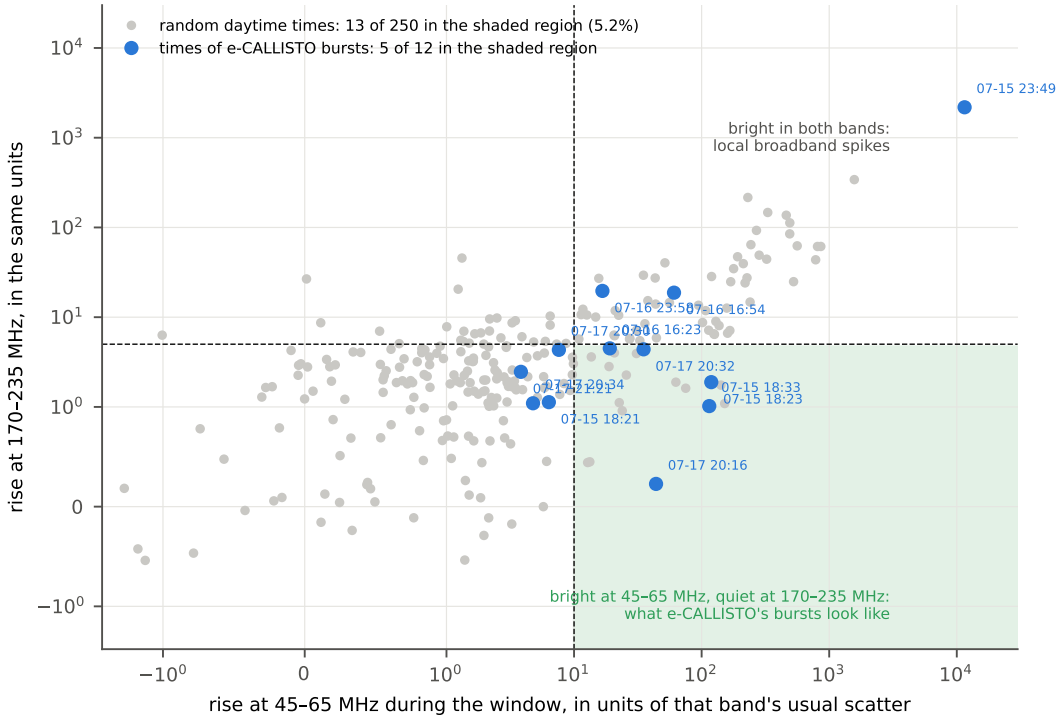


Figure 8: The burst test. Each point is one 15–40 s window of raw box-air data. Its position says how much the raw power rose during the window, at 45–65 MHz (across) and at 170–235 MHz (up), in units of that band's usual scatter. Blue: the windows of the e-CALLISTO bursts, labelled by start time. Grey: 250 random windows in the same daytime hours. The bursts e-CALLISTO recorded are bright at low frequency and faint above 170 MHz, so a real one should land in the shaded region ($z_{45–65} > 10$, $z_{170–235} < 5$). The local spikes of §6.6 rise in both bands and land at the upper right. 5 of 12 bursts fall in the shaded region, against 5.2% of the random windows. Axes are linear between −1 and 1 and logarithmic beyond.

6.6 Impulsive events in the raw data are local

Looked at without a reference list, raw box-air is full of impulsive events: single integrations that rise at every frequency at once, about 126 times an hour in daytime (Fig. 9, as horizontal lines). They do not drift. `impulsive_scan.py` finds them in all of phase C (44 h of antenna data) and sorts them by whether the Sun is above the ridge at the antenna, up but behind the ridge, or down. Their rate follows the time of day, not the Sun: 126 per hour with the Sun visible, 123 with it behind the ridge, and 8 at night (24, 29 and 0.8 for events lasting 3 or more integrations). If they were solar, the ridge would hide them. They are local, from a daytime source at the site we have not identified, and they are why a search for bursts must start from an independent list.

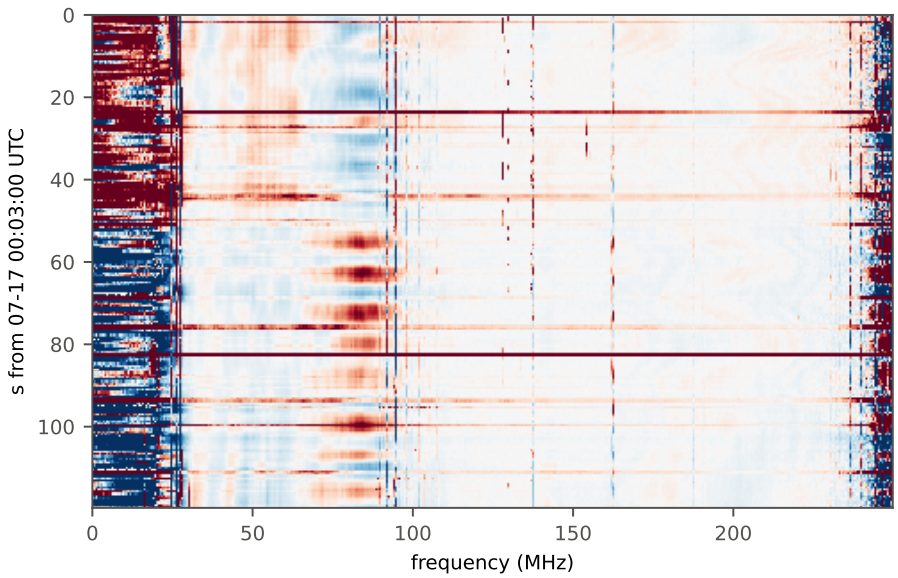


Figure 9: Two minutes of raw box-air data in daytime with the Sun behind the ridge, each channel divided by its median, time running down. The horizontal lines across all frequencies are the impulsive events.

6.7 Sunrise and sunset in the raw data

The calibrated data hold one terrain crossing. Phase C holds 6, the rest uncalibrated (Table 3). `sun_transitions.py` keeps the raw context of each, minute by minute, ±60 min (Figs. 11 and 12): the mean raw antenna spectrum, whether the transmitter was on, whether the antenna moved, and how many rows were antenna rows rather than calibration states.

It then fits raw power at each beam frequency as a line plus the Sun column at that crossing's pointing, over the parked stretch of data containing the crossing, without transmitter-on minutes unless the whole stretch has it on. Raw power has no kelvin scale. The fit gives $S_\odot/T_{\rm sys}$, and the 5th–95th percentile of the calibrated temperatures of 07-17 stands in for $T_{\rm sys}$. That stand-in comes from other hours and heights, so the fluxes in Table 7 are good to a factor of about 2.

The crossings in context. At these pointings the Sun's predicted signature in raw power is not a step but a slow bump. As the Sun sinks towards the ridge, the beam's gain toward it changes, and the ridge then cuts it off over 10–30 min. So it peaks some 20–30 min before sunset (the sunrise version is the blue curve of Fig. 4). It amounts to a few percent of the power for a Sun of a few SFU. Figures 11 (sunsets) and 12 (sunrises) show what the raw data do.

- **07-15 sunrise, 30 m**: there are no antenna rows from 2 min before the crossing to 20 min after it; no fit.
- **07-16 sunset, 30 m, transmitter off**: the quietest crossing. Raw power rises steadily, by about 10% over the two hours, at all four frequencies. The fit puts a Sun of 8–11 SFU at 199 MHz on top of that

rise, mostly from a bump 15–30 min before sunset. The bump is no larger than the unexplained wander elsewhere in the window.

- **07-16 sunrise, 30 m**: box-air's self-EMI comb (memo 006) is on until 16:51, and its lines cover much of the raw spectrum; the fitted fluxes are erratic.
- **07-17 sunset, 30 m**: the antenna is parked about 73° below horizontal, but its elevation wanders by several degrees within each minute (black ticks) until it steps at 00:02:45 and settles near −66°. The raw power below 100 MHz is dominated by a solar radio burst that ends as the Sun goes behind the ridge (next paragraph). No stretch with steady pointing spans the crossing, so no fit.
- **07-17 sunrise, 87.5 m**: the transmitter is on for most of 15:36–16:14 (orange ticks), and its leakage dominates those minutes (the 10–30% excursions in the bottom panel). Calibration cycles also interrupt the antenna rows (grey). On transmitter-off minutes, including ones after the calibrated window ends at 16:22, the raw power is flat to about 1%, and the Sun term is 11–16 SFU at 199 MHz.
- **07-18 sunset, 91 m, transmitter on throughout**: fitted on channels midway between the comb teeth. The raw power wanders by several percent, and the Sun term is 7–10 SFU at 199 MHz.

A solar burst at the 07-17 sunset. `validate_sunset_burst.py` reads the raw spectra of both boxes from 23:44 to 00:08 UTC and divides them by their median over the quiet minutes on either side (Fig. 10). From 23:53:45, box-air's power below 100 MHz rises far above that. Over 23:55:30–00:00:00 it is 1.09 times the reference at 50–60 MHz, 1.54 at 66–72 and 2.30 at 78–84 MHz, and 1.18 at 100–120 and 0.99 at 180–230 MHz.

- **It is not interference.** It is smooth across the band, and channels at whole megahertz rise like their neighbours (1.58 against 1.56), so it is not box-air's 1 MHz self-EMI comb (memo 006). Box-gnd, on the same correlator, stays at 0.95–1.05 times its reference below 90 MHz. The excess's lower edge drifts from about 80 to 50 MHz over ten minutes, with a fainter band near 160–175 MHz.
- **It is not pointing.** At 72–84 MHz it peaks at 2.5 times the reference at 23:58, at an elevation of −73°. At 23:44, at the same elevation, the power was 0.97 times the reference.
- **It ends as the Sun goes behind the ridge.** For box-air the Sun's disk is covered between 00:00:23 and 00:02:45 (centre at 00:01:34). In 10 s means, the excess falls below 1.5 times the reference at 00:00:45 and below 1 at 00:02:35, while the elevation is still wandering about its park value of −73°. The elevation then steps at 00:02:45 and is within 1° of its final −66° from 00:03:25. With the step the power falls by a further 18%.
- **A flare was in progress.** GOES-18 recorded a C1.5 X-ray flare from 23:51 to 00:09 UTC, peaking at 00:02 (SWPC event 7990, `inputs/solar/20260716events.txt`). The same event has a 245 MHz burst of 820 SFU at 00:07–00:08 (Palehua), after the Sun had set for box-air; box-air at 180–230 MHz stays within 4% of its reference from a minute before it to two minutes after. e-CALLISTO stations (HAARP, COHOE) recorded a type III burst at 23:58:59 (34 s, 20–75 MHz) inside the excess (Table 6).

So the excess is solar emission during that flare, cut off for box-air by the ridge. Its slow downward drift and the band near twice its frequency resemble a type II burst, but no observatory's sweep is listed to confirm the type. That box-gnd misses it would fit a Sun already below box-gnd's own horizon, but box-gnd's position is not solved. This replaces the earlier reading of the crossing, which tied the step in raw power to the elevation settling. Most of the fall comes before the elevation moves, while the disk is being covered; only the last 18% comes with the elevation step.

What the crossings say. Over the 3 fitted crossings without the self-EMI and 121–219 MHz (where above 2σ), the Sun term runs from 3 to 16 SFU. The concurrent prediction (RSTN at 245 MHz with the MWA quiet-Sun shape, Fig. 5) is 2–11 SFU over the same frequencies. RSTN's noon flux at 245 MHz stayed between 13 and 17 SFU from 07-15 to 07-18. So the raw data allow a Sun of the expected size. They do not detect it convincingly: the Sun term is a few-percent bump on raw power that drifts by as much for other reasons, and the kelvin stand-in is good only to a factor of 2. What they do show is that no crossing has a jump as large as the calibrated 07-17 excess implies, which is 2.0–3.2 times larger than their upper bounds (32 SFU at 199 MHz). That includes the same 07-17 crossing in raw data: with a longer stretch of transmitter-off minutes, it gives 11–16 SFU at 199 MHz.

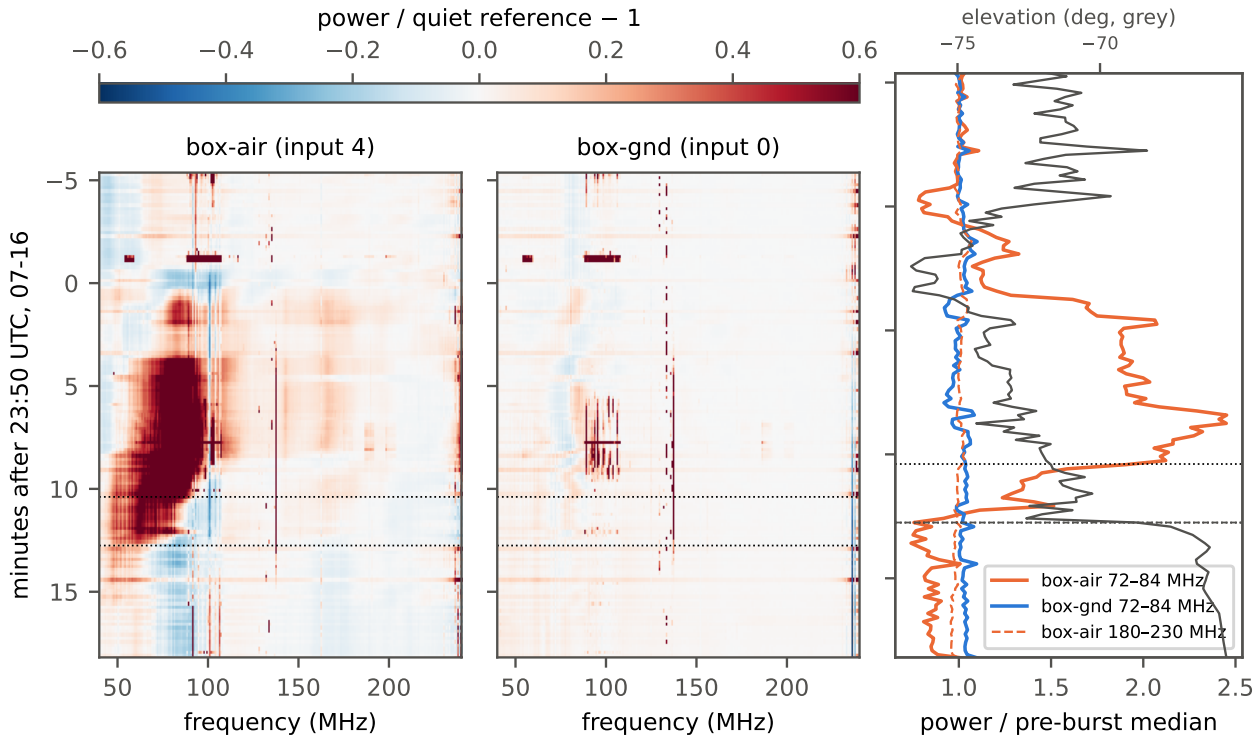


Figure 10: The 07-17 sunset at 30 m in raw spectra (`validate_sunset_burst.npz`; raw correlator files, all integrations on the antenna; 10 s means). Left and middle: box-air and box-gnd power divided by their median over quiet minutes before 23:53 and after 00:03 UTC, minus 1, with time running down. Right: band power over its pre-burst median for box-air at 72–84 MHz (solid orange) and 180–230 MHz (dashed), and box-gnd at 72–84 MHz (blue); the antenna's elevation in grey (top axis). Dotted lines: the Sun's disk going behind the ridge for box-air. Grey dashed: the elevation steps away from its park value. The excess is box-air's alone, drifts downward in frequency, and ends while the disk is being covered.

The timing is weakly constrained. The best shifts τ^* in Table 7 range from -60 to -4 min, with broad χ^2 minima; 1 of them sits at the ± 60 min edge of the scan, so its shift is not bounded. The ground under the antenna goes into shadow about half an hour before the antenna at the 30 m sunsets and nearly two hours before at the 91 m one, and comes into sunlight 15–50 min after it at the sunrises ("ground" column). A signal from the lit ground, or from heating there, would therefore lead at sunset and lag at sunrise. These data cannot tell those timings apart.

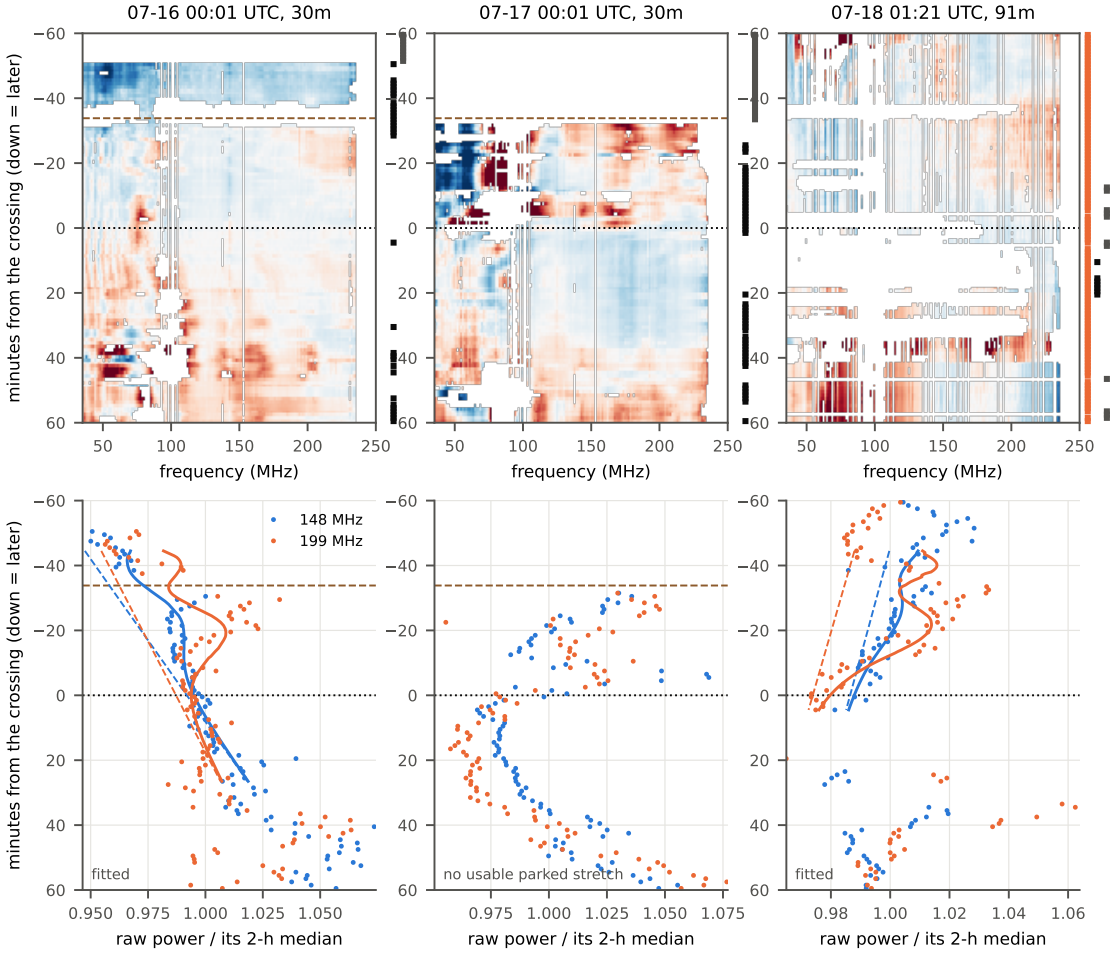


Figure 11: The three terrain sunsets in raw box-air data, one column each, time running down; the dotted line is the Sun's centre crossing the ridge at the antenna. Top: the raw spectrum, minute by minute, divided by its median over the two hours. Ticks at the right mark minutes with the transmitter on (orange), the antenna moving (black) or calibration rows (grey); the brown dashed line is when the ground under the antenna goes into shadow. Bottom: raw power at 148 and 199 MHz, divided by its median, on the same time axis. Where a fit was made, the dashed line is its baseline and the solid line baseline plus Sun. If the Sun is seen, the points follow the solid line, not the dashed one, through the half hour before the crossing.

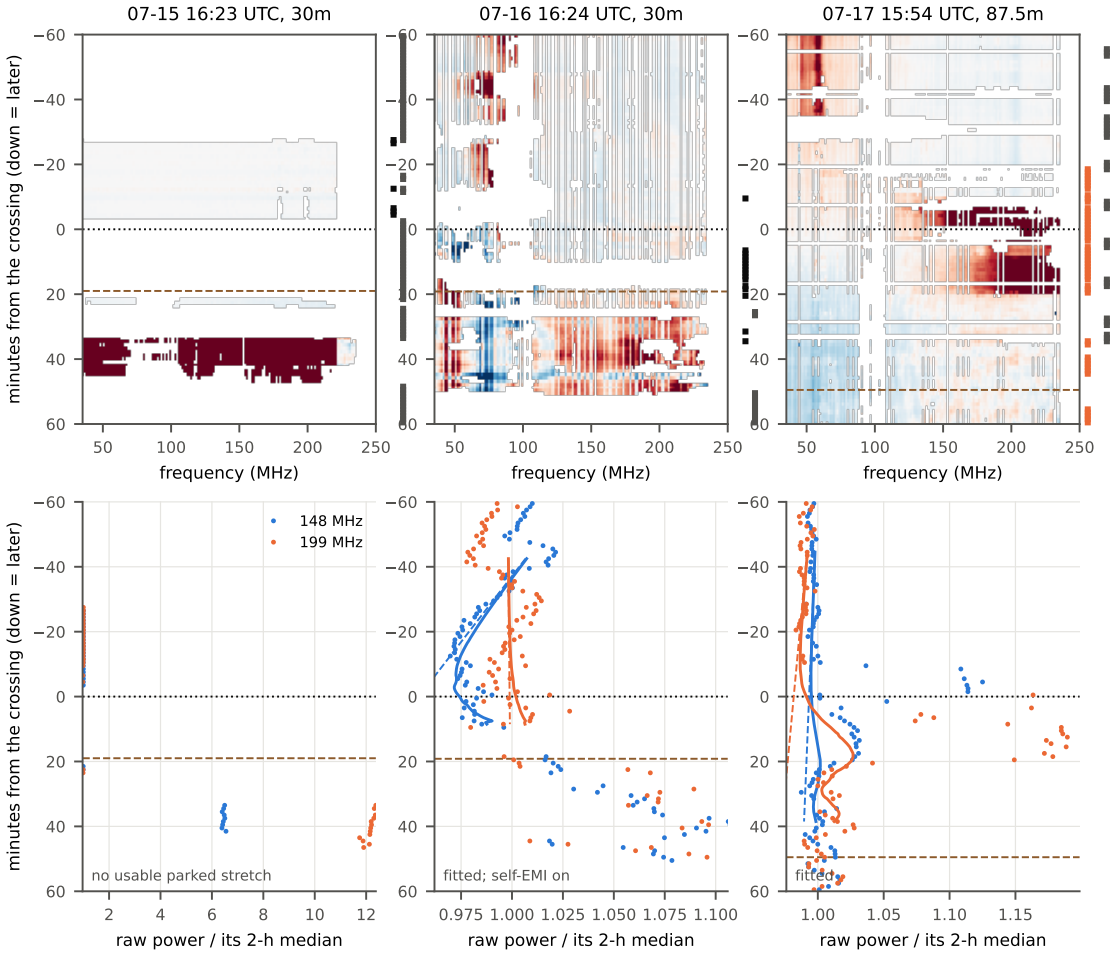


Figure 12: The three terrain sunrises, as in Fig. 11; here the brown dashed line is when the ground under the antenna comes into sunlight, and the Sun's signature follows the crossing instead of preceding it.

crossing	status	fit (min)	τ^* (min)	ground (min)	121	148	199	219
					$S_\odot$ (SFU, range from T_{sys} stand-in)			
rise 07-15 16:23 (30m)	no usable parked stretch	–	–	+19	–	–	–	–
set 07-16 00:01 (30m)	fitted	-44 to +26	-4	-34	5–8	4–7	8–11	7–13
rise 07-16 16:24 (30m)	fitted; self-EMI on	-42 to +10	-16	+19	1–1†	19–32	3–5†	6–10
set 07-17 00:01 (30m)	no usable parked stretch	–	–	-34	–	–	–	–
rise 07-17 15:54 (87.5m)	fitted	-44 to +38	-4	+50	3–5	3–4	11–16	1–2†
set 07-18 01:21 (91m)	fitted	-44 to +10	-60	-114	8–12	4–7	7–10	4–7

Table 7: Raw-power fits at the terrain crossings (`transitions_v0004`): status, the fitted stretch and best timing shift τ^* (minutes from the crossing), when the ground under the antenna crosses ("ground", minutes), and $S_\odot$ at four frequencies (MHz) for the 5th–95th percentile of the 07-17 calibrated temperatures. †: below 2σ.

6.8 The Sun's path through the beam

Figure 13 shows the Sun column, minute by minute over phase C (`sun_profile_v0004`, 2899 minutes with pointing). Parked, it changes slowly with the Sun's motion, between the terrain crossings. Moving, it changes fast: during the 07-17 raster the beam's gain toward the Sun swings by factors of several within seconds. On 07-15 and 07-16 much of the day has no valid pointing in the table, and outside the raster the table's azimuth zero inherits potentiometer slips (memo 003). So the predicted Sun there is uncertain.

The raster offers a test with no kelvin scale. Since the receiver gain multiplies everything, regressing raw power on the sky (GSM through the beam), the ground, a constant, the transmitter's leakage, a drift and the Sun column gives the Sun's flux on GSM's brightness scale (`raster_sun.py`). The null is the same fit with fake Suns at the same altitude in 11 other azimuths. The test fails (Table 8). The real Sun beats every fake at 4 of 13 frequencies with the HFSS beam and 10 of 25 with the empirical beam, and the fitted fluxes change sign (8 and 20 negative). The residual, 1.9% of the power with HFSS, depends on direction and is larger than the Sun's modulation. Any extra direction-dependent column absorbs it.

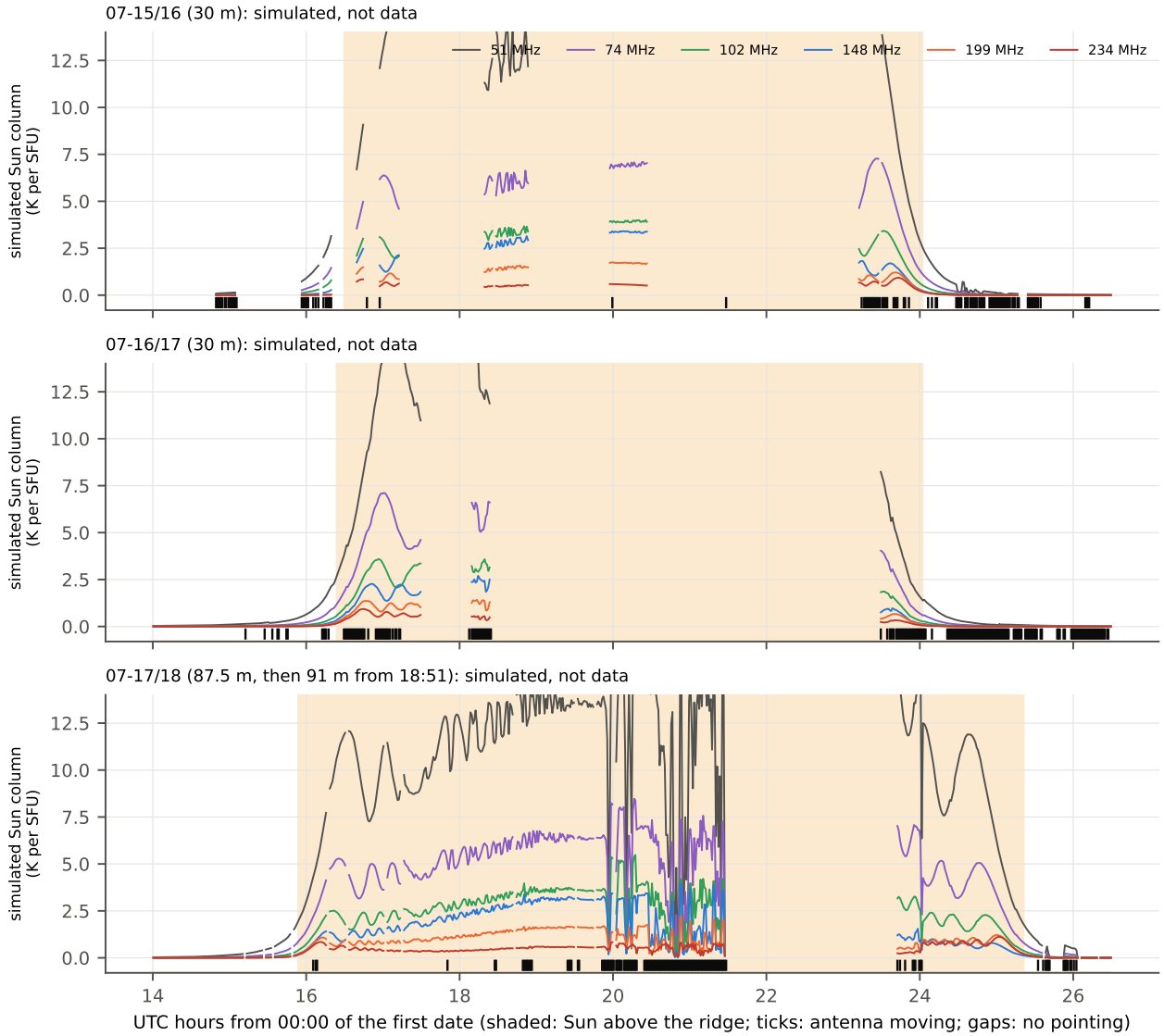


Figure 13: Simulated, not data: the predicted Sun column (antenna temperature per SFU of solar flux) through box-air's HFSS beam over phase C, at six frequencies, from the Sun's position, the antenna's recorded pointing and the knife-edge terrain visibility (`sun_profile_v0004`). No measured power is plotted.

	HFSS beam				empirical beam	
MHz	$S_\odot$ (SFU)	fakes $\geq$	resid.	MHz	$S_\odot$ (SFU)	fakes $\geq$
62.5	-3	3/11	1.3%	64.5	-0	10/11
74.2	-2	7/11	1.2%	70.3	-4	2/11
113.3	-11	2/11	1.9%	76.2	-1	8/11
125.0	-8	5/11	1.7%	82.0	-5	2/11
136.7	-19	5/11	2.0%	113.3	-12	2/11
148.4	-125	3/11	1.6%	119.1	-9	0/11
160.2	12	5/11	1.8%	125.0	-8	7/11
171.9	21	3/11	2.5%	132.8	-14	3/11
183.6	38	0/11	2.1%	138.7	-19	5/11
195.3	-40	0/11	1.8%	144.5	-47	5/11
207.0	12	0/11	2.6%	150.4	-74	3/11
218.8	-35951	0/11	4.3%	156.2	106	4/11
230.5	14	1/11	8.8%	162.1	9	6/11
				168.0	39	1/11
				173.8	16	5/11
				179.7	-117	0/11
				185.5	85	0/11
				191.4	-150	0/11
				197.3	-40	0/11
				203.1	-17	0/11
				209.0	-69	0/11
				214.8	-42	0/11
				220.7	-14	0/11
				226.6	-27	0/11
				232.4	-7	5/11

Table 8: The raster test (`raster_sun_v0004`): the Sun's flux on GSM's scale, how many of the 11 fake Suns improve χ^2 as much or more, and the residual rms as a fraction of the power.

6.9 What this does and does not show

- **Box-air receives solar emission.** Solar radio bursts recorded by independent spectrographs appear in raw box-air data at the same seconds, with the type III drift and the low-frequency-weighted spectrum those stations saw (§6.5). That is a physical identification, not a coincidence in timing.
- **The raw crossings allow a quiet Sun of the expected size but do not detect it.** Across 3 uncalibrated terrain crossings on three days and three heights, the Sun term is 3–16 SFU at 121–219 MHz, against a concurrent prediction of 2–11 SFU (§6.7). It is a few-percent bump on raw power that drifts by as much for other reasons, and its kelvin stand-in is good only to a factor of 2.
- **The calibrated 07-17 excess is larger than the Sun.** At 15:54 the calibrated spectrum jumps by 3.6 times the concurrent prediction (1.4 times scaled by a), and the local fit gives 2.0–3.2 times the other crossings at 199 MHz. RSTN shows no brightening that day. Part of that excess is probably not the Sun. Anything that changes as the antenna and its box come into sunlight follows the same geometry, so it is not separated. The excess rests on 7 one-minute bins at the end of the calibrated window, all directly after the transmitter switches off.
- **The timing does not separate the Sun from the lit ground.** Best timing shifts range from -60 to -4 min, with broad minima, and 1 sits at the edge of the ± 60 min scan.
- **The beam's sweep during the raster does not isolate the Sun** with the present sky/ground/beam model: fake Suns fit as well (§6.8).
- $S_\odot$ inherits the unresolved factor $a \approx 2$ and HFSS's gain at $\sim 50°$ from boresight.

7 Recovering the sky and the ground

The night fit (§5) holds one pointing at one height, so the ground temperature and the offset are degenerate there. This section tries three ways round that, each with a check that could fail.

7.1 Tying the frequencies together

At one pointing and height, the beam's ground fraction $f_{\mathrm{gnd}}(\nu)$ still changes with frequency. If the ground is one temperature at all frequencies (thermal emission) and the receiver offset is a smooth function of frequency,

the pedestal $f_{\rm gnd}(\nu)T_{\rm gnd} + T_{\rm off}(\nu)$ could separate them. `joint_fit.py` reduces each frequency of the night fit to a likelihood for $(T_{\rm gnd}, T_{\rm off})$, marginalizing the sky and the GSM amplitude. It then fits one $T_{\rm gnd}$ and a Legendre polynomial in $\log\nu$ for the offset over 105–235 MHz. If the data fixed $T_{\rm gnd}$, it would not move with the polynomial's order, nor between the halves of the band. It does: 275 to 920 K as the order goes from 1 to 6, and the two halves disagree by hundreds of kelvin (Table 9). The ground fraction's frequency dependence is too much like a smooth offset; smoothness alone does not fix $T_{\rm gnd}$.

offset order	$T_{\rm gnd}$ (K), all	lower half	upper half
1	438 ± 59	529 ± 239	-173 ± 249
2	400 ± 132	457 ± 247	-193 ± 250
3	275 ± 136	533 ± 252	899 ± 282
4	641 ± 176	566 ± 253	927 ± 282
5	920 ± 187	566 ± 253	933 ± 282
6	820 ± 188	567 ± 253	937 ± 282

Table 9: $T_{\rm gnd}$ from the night fit tied across frequency (`joint_v0004`), against the order of the offset's polynomial, over the whole band and its two halves.

7.2 The sky as calibrator on parked nights

Most nights have no kelvin scale, but the Galaxy drifts through the beam. In a stretch of fixed pointing, raw power at each frequency is fitted as gain × (GSM through the beam) + a constant + a linear drift (`sky_cal.py`). The gain is then in raw units per GSM kelvin. The constant over the gain is $T_{\rm rx} + f_{\rm gnd}T_{\rm gnd}$ in GSM kelvin, and stretches at different heights, with different $f_{\rm gnd}$, would separate the two. The receiver regime is rx-A on all three nights. A stretch enters that comparison only if the sky fits it, with a median residual under 5% of the power. Figure 14 shows the raw data with the fit.

- **07-17, 87.5 m** (calibrated): the sky fits to 0.5% rms. The gain against `tcal`'s gives a =1.8–2.1 at 105–220 MHz, as the night fit found. This checks the pipeline, not where a comes from.
- **07-16, 30 m**: box-air's self-EMI is on all night; with its teeth masked the sky fits to 3.8%.
- **07-15, 30 m**: the switch daemon was off, so the rows carry no switch state. Before about 04:45 the raw power is a few percent of its later level, and a fit over the whole night misses the sky badly. Fitted only from 05:12 UTC, a cut chosen after looking at these data, the sky fits to 1.9%. The table has no valid pointing after 05:12: all of the fitted bins (100%) take the pointing of the two earlier bins that have any, which assumes the antenna stayed parked all night.

That gives 3 nights, whose ground fractions span only 0.05. A 10 K difference between the nights in anything else then shifts $T_{\rm gnd}$ by about 191 K, and the fit gives −2734 to 2506 K at 120–180 MHz. Parked nights at these two heights cannot separate the ground, with or without 07-15.

7.3 The ground from the rotating antenna

When the antenna moves, the beam's ground fraction swings between about 0.5 and 0.75 within seconds, while the receiver gain stays put. In five moving windows (`raster_sky.py`), raw power per integration is regressed on GSM through the beam, the ground fraction, a constant, the transmitter's leakage, the Sun and a drift. The gain multiplies every column, so the ground coefficient over the GSM coefficient is $T_{\rm gnd}$ in GSM kelvin, with no kelvin calibration. Errors are jackknifed over 2-minute blocks.

The excess in the raw data. Figure 15 shows it in the raster's raw power, integration by integration, as the beam's ground fraction swings between 0.60 and 0.75 over each elevation sweep. Take the raw power with no RFI flags at all, and subtract every fitted term except the ground: the sky, the constant, the transmitter, the Sun and the drift. What is left at 64 and 82 MHz rises and falls with the ground fraction by 34 and 44% of the power, peak to peak. A 300 K ground would move it by a few percent. The slope of that residual against the ground fraction is the ground's brightness: 2404 and 1628 K in GSM kelvin with no flags, and 3620 and 1797 K on the RFI-flagged samples (18 and 21% of them). So at low frequency the excess is in the raw data, and the

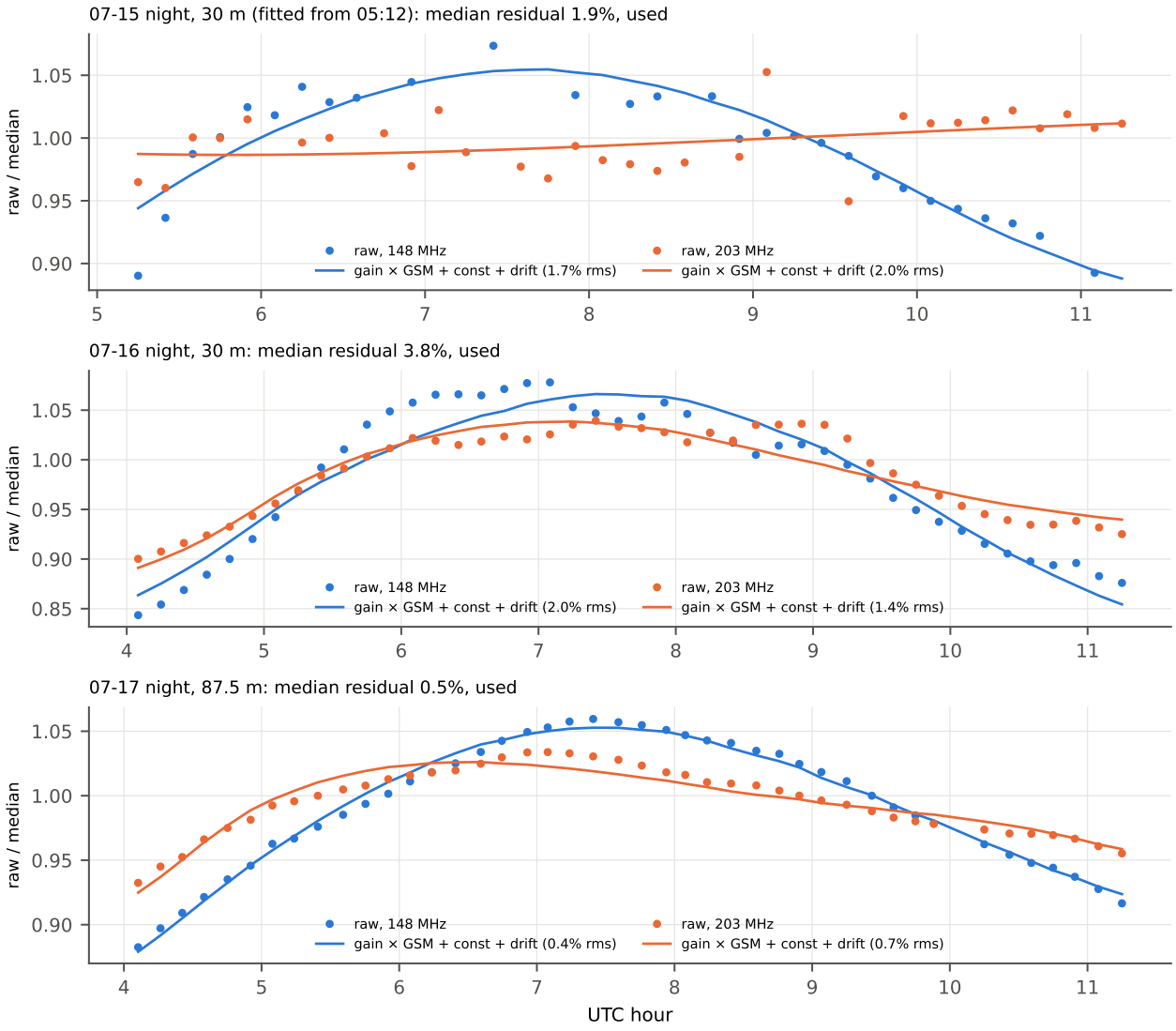


Figure 14: The three parked nights in raw data at 148 and 203 MHz, each divided by its median (points), with the fitted gain × GSM through the beam + constant + drift (lines) (`skycal_v0005`, HFSS beam). 07-15 is fitted only from 05:12 UTC (see text).

flags change its size but did not create it. At 162 MHz it is small and depends on the flags: 134 K unflagged and 236 K flagged (51% kept). At 113 MHz the unflagged raster is dominated by the transmitter's leakage, so only flagged values are quoted there. The flags keep fewer than a third of the raster's samples below 120 MHz, and the fraction kept depends on where the beam points. The flagged values above 100 MHz therefore carry a selection that the unflagged ones do not.

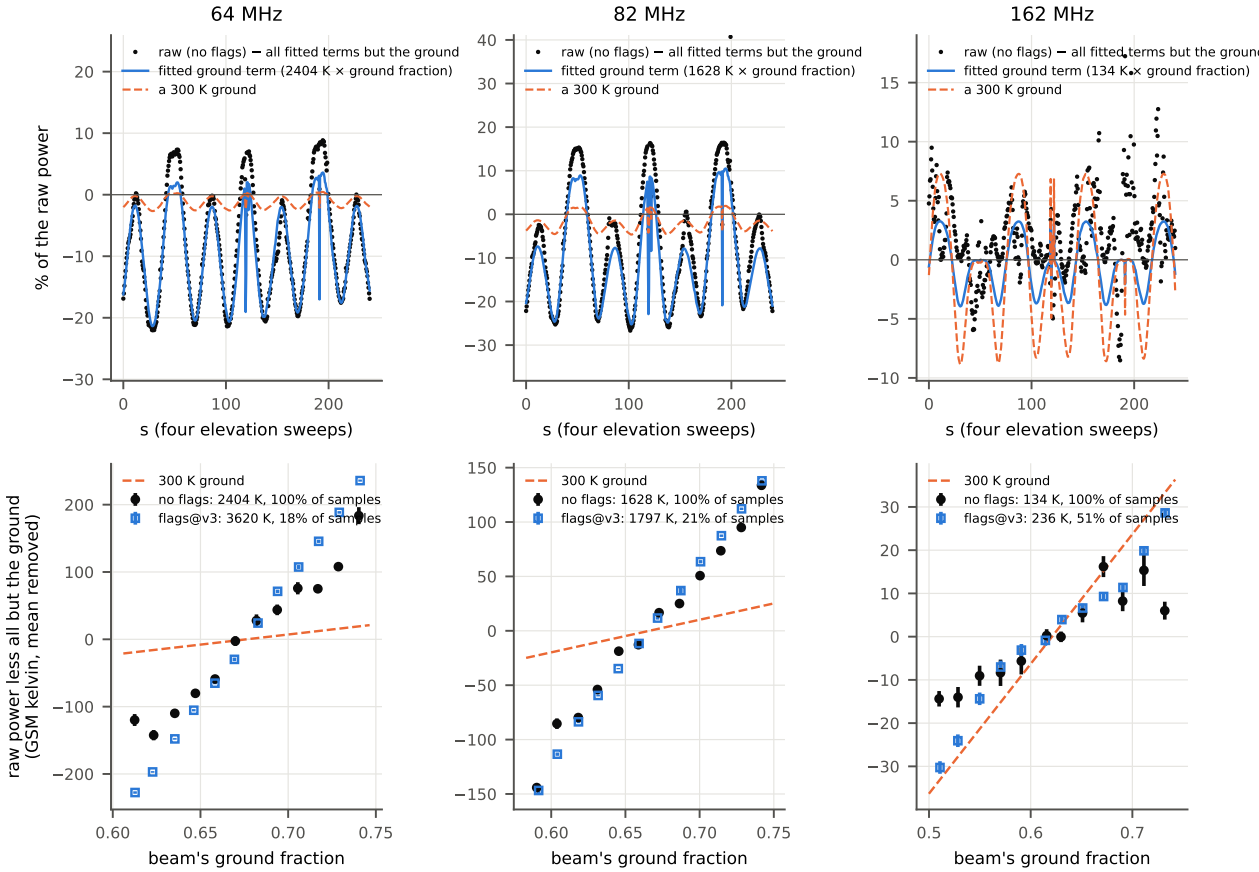


Figure 15: The ground excess in the raster's raw power (`raster_sun_v0004`, raw correlator files for the unflagged values; channels midway between the transmitter's comb teeth). Top: four elevation sweeps of raw power with no RFI flags, less every fitted term except the ground (points), against the fitted ground term (blue) and what a 300 K ground would give (orange), in percent of the power. Bottom: the same residual, binned by the beam's ground fraction and converted to GSM kelvin, with no flags (filled) and with `flags@v3` (open); the slope is the ground's brightness.

Figure 16 and Table 10 show the result in every window.

- **It repeats.** The three 91 m windows (two evenings, the transmitter and the Sun in different states) agree at 9 frequencies between 120 and 180 MHz (χ^2/dof up to 6.2).
- **It is not flat in frequency.** It falls from 490 K at 125 MHz and 336 K at 139 MHz to 195–207 K at 150–175 MHz. Below 85 MHz every window gives thousands of kelvin: 1970–3620 K at 64 MHz.
- **The 30 m windows sit lower and scatter more**; the 07-16 one has the self-EMI on.

Thermal ground emission should not depend on frequency. At 150–175 MHz the values are those of soil at about 290 K with an emissivity near 0.7, but that coincidence does not identify them. What rises below 140 MHz is not the ground's temperature. It enters wherever the beam sees the ground, and it grows steeply toward low frequency like the sky.

Not reflected sky. Ground reflects part of the sky, which is far brighter at low frequency. A specularly reflected sky term (each below-horizon direction mirrored to the sky above, flat ground) was added in the raster, with the ground's reflectivity R free. Soil would give $R \approx 0.1$–0.4. At 64–82 MHz the fit gives $R = -0.15$ to -0.03, and the ground's emission term stays at 1831–3704 K. Specular reflection does not explain the excess, and diffuse reflection is tested over phase C in §7.5. Candidates left are the beam below the horizon (both beams are extrapolated there; the raster does not sample it, memo 005) and something fixed to the platform that turns with the antenna.

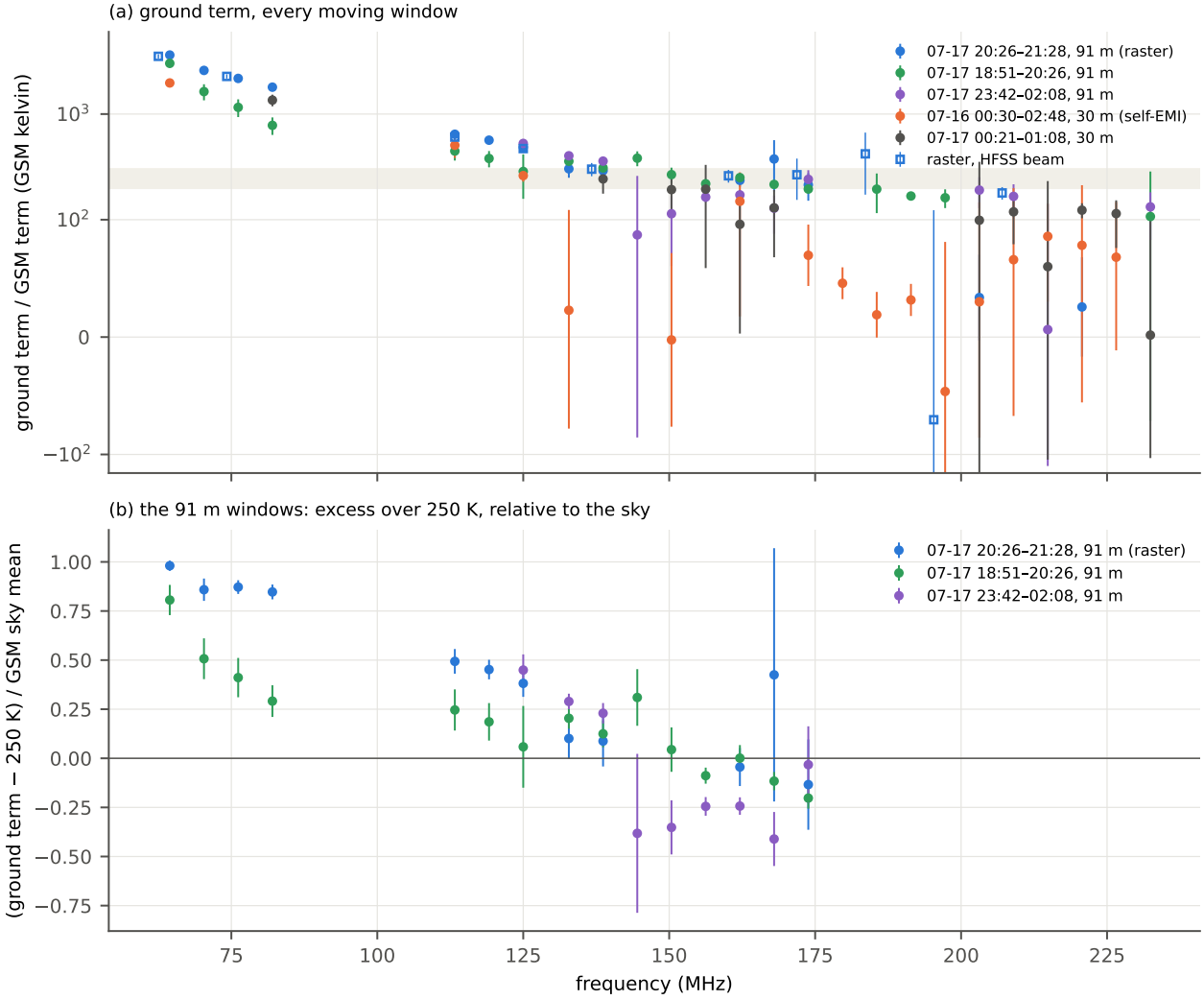


Figure 16: (a) The ground term in each moving window, as the ratio of the ground and GSM coefficients (GSM kelvin; empirical beam unless marked; jackknife errors; points with errors above 300 K omitted). Shaded: 200–300 K. The axis is linear below 100 K and logarithmic above. (b) For the three 91 m windows, the same ground term less a thermal 250 K, divided by GSM's whole-sky mean (below 180 MHz). A sky leaking in with a fixed fraction would give a constant.

7.4 Tracing the excess

Four explanations of the excess make different predictions, and the rotating-antenna data can test three of them.

Where on the ground. `ground_excess.py` repeats the regression with the ground split into regions, each with its own brightness (Table 11).

MHz	mean of three 91 m windows (GSM K)	χ^2/dof
125.0	490 ± 31	3.1/2
132.8	379 ± 16	3.9/2
138.7	336 ± 19	2.1/2
144.5	349 ± 58	2.6/1
150.4	205 ± 34	4.9/1
156.2	195 ± 11	6.2/1
162.1	201 ± 11	10.7/2
168.0	207 ± 13	4.9/2
173.8	199 ± 14	0.8/2

Table 10: The ground term, weighted mean of the three 91 m windows, where at least two have errors under 200 K, and the windows' agreement.

- **Horizon band against deeper ground.** Radio noise from the landscape, man-made or lightning, would arrive from just below the horizon. In the raster the band within 10° below the horizon and the deeper ground carry similar excesses with the flags. With no flags the band is somewhat brighter, but the deep ground still carries thousands of kelvin at 64–84 MHz. The deep ground, most of the beam's ground weight, is not reached by noise from the horizon.
- **Azimuth.** Sources on the landscape would sit at particular azimuths. In the raster, eight azimuth sectors give 1767–2542 K at 76 MHz (typical error 311 K): the excess covers the ground evenly. The other windows sample too few orientations to constrain eight sectors.

Against the sky. A sky seen where the model expects ground would give an excess proportional to the sky's brightness. That covers sky scattered diffusely by rough ground, or a back lobe stronger than modelled. Figure 16b divides the excess (the ground term less 250 K) by GSM's whole-sky mean. In the raster the ratio is 0.85–0.98 at 64–82 MHz and falls to near zero by 150–175 MHz (within ±0.4). It also differs between windows. Its spectral index between 64 and 82 MHz is 3.2, steeper than the sky's (≈2.5). A constant scattering fraction or back-lobe gain does not reproduce this; a coupling to the sky that weakens fast with frequency would.

A gain that follows the orientation. If the system gain or the antenna's match changes as the antenna turns, by an amount that follows the ground fraction, the change multiplies a nearly constant system power. It then has the same shape as the ground column and cannot be told apart from it by these regressions. A gain swing as large as the ground term's swing over a sweep (Fig. 15: 34% of the power at 64 MHz, 44% at 82 MHz, 8% at 162 MHz, peak to peak) would make the whole excess. A gain change that multiplies only the sky (an extra column of ground fraction times GSM) does not absorb it: at 64–82 MHz the ground term stays at 2372–3712 K, against 1797–3620 K without it. Testing an orientation-dependent match needs the antenna's reflection coefficient at different tilts. All 19 field antenna-S11 sweeps from 07-13 on were taken upright: those with pointing (19) are within 3.2° of el = 0. So the data cannot test it.

So far. The excess is not specular reflection of the sky, not noise arriving from the horizon, and not concentrated in any azimuth. It does not scale with the sky as a fixed fraction. What is left is a coupling that falls steeply with frequency and follows the antenna's orientation relative to the ground: either the beam's response below the horizon, which both beam models extrapolate, or the antenna's match or gain changing as it tilts. Telling them apart needs a measurement, not more regression: S11 or a calibration cycle with the antenna tilted, or a beam measurement that samples directions below the horizon. The daily term over phase C (§7.5) is a separate test, with the sky drifting under fixed pointings.

7.5 The low-frequency daily term over phase C

The moving windows each last about an hour, so the sky hardly changes within one and cannot be told from a fixed ground term. Phase C adds what they lack: three days of sidereal time. `global_fit.py` models rx-A raw power in `raw_binned_v0001` over all of phase C (6816 bins in 46 gain windows that pass its residual gate), one frequency at a time. The model is a smooth gain times [GSM through the HFSS beam + a ground term

window, region	64	70	76	82	113	125
raster, flags@v3: within 10° below horizon	4175±236	3019±496	2379±261	1637±239	970±140	920±340
raster, flags@v3: deeper ground	3574±95	2613±169	2179±83	1793±73	653±48	510±51
raster, no flags: within 10° below horizon	5444±1046	3048±1062	3312±647	2491±456	–	293±314
raster, no flags: deeper ground	2671±791	1884±158	1818±269	1641±134	-3±805	398±45
07-17/18 night, flags@v3: within 10° below horizon	-1923±746	–	–	–	–	-351±232
07-17/18 night, flags@v3: deeper ground	3485±1111	–	1138±932	2104±898	783±175	535±24
raster, 64 MHz, 8 azimuth sectors	4079±957, 3070±725, 5100±1432, 4422±1588, 3279±527, 5644±1089, 2061±599, 4008±549					
raster, 76 MHz, 8 azimuth sectors	2300±272, 2180±329, 2177±316, 1767±232, 2542±337, 1810±236, 2312±315, 2323±306					

Table 11: The ground split into regions (`ground_excess_v0004`, empirical beam; brightness in GSM kelvin, jackknife errors; "–": error above 2000 K). Top: the band within 10° below the terrain horizon and the deeper ground, at six frequencies (MHz). Bottom: the raster's eight azimuth sectors (0–45°, 45–90°, . . . , east of north).

+ an offset]. The log gain is piecewise linear in time, with knots every 3 h and a 5% prior on each knot step. There are no sky corrections ($l = 0$). Alternate 30 min blocks are held out, and every model is scored by the held-out rms as a fraction of the power. The code is on the data-analysis branch `daily-term-fast-gain-v0004` (commit ad209ab), pushed and not merged; it extends `diffuse-ground` and `daily-term-fast-gain`, which hold the same code on terrain v0003. `validate_phase_c.py` in this directory runs each configuration on the 12 beam frequencies of the original analysis, 62.5–175.8 MHz; the 6 from 62.5 to 82.0 MHz are reported. The ground term takes one of these forms, each with one free parameter unless stated:

- a constant temperature T_{g};
- diffuse (Lambertian) reflection of the sky, $(1 - R)\,290\,\mathrm{K} + R\,\bar{S}(t)$. Here $\bar{S}$ is the cosine-weighted GSM mean over the open sky above the era's horizon, 1153–5184 K at these frequencies, so a ground term of thousands of kelvin is the size this form would give;
- the same, with $\bar{S}$ taken 6 h earlier, 6 h later or 12 h away in sidereal time, as controls;
- "beam": the beam's response below the horizon falling on the sky of the current pointing instead of on the ground;
- "pickup": $k\,\bar{S}$ without the ground fraction, for sky entering by a path that does not turn with the antenna;
- a housekeeping stream, averaged over each bin from the correlator metadata: three RF-switch thermistors, the load temperature and the system current;
- "free level": $T_0 + R\,\bar{S}$ with both free and a 290±300 K prior on T_0.

Over all of phase C (Table 13), reflection with the real sky beats a constant ground by 0.25–0.32 points of held-out rms, with $R = 0.60$–0.83. The shifted skies gain at most 0.27 points. The beam form differs from a constant ground by at most 0.03 points, so the term does not follow the sky of the current pointing. Pickup does as well as reflection or slightly better (by 0.01–0.12 points, with 2 of its fits diverged), so the ground fraction is not what carries the term.

Not a recorded temperature or current. On the 93% of bins with all five streams, the best single stream does 0.18–0.31 points worse than reflection at 62–78 MHz; at 82 MHz the first thermistor does better (5.54 against 5.85%). With all five streams in the fit at once, reflection with the real sky still lowers χ^2/dof by 8.2–21.1, against at most 1.8 for the shifted skies, and R stays 0.61–0.70 at 62–74 MHz.

The controls need the tied level. The reflection form ties the ground's constant part to R. With the level free, the sky shifted by 6 h beats the tied real sky by 0.16–0.43 points at 62–70 MHz, with R near -1, which is unphysical. At 74–82 MHz the tied real sky still beats every shifted free-level fit, by 0.08–0.17 points. The free real sky itself, kept from diverging by the prior (0 of 6 fits flagged), gives $R = 2.4$–6.9 with T_0 from -11 to -5 kK: far more time variation than reflection allows, offset by a negative level.

Disjoint subsets (Table 14). Splitting phase C at its median fitted time, 07-16 18:10 UTC, gives halves A and B. In half A the reflection form is the best model at 5 of 6 frequencies, and the 6 h-shifted sky does 1.0–1.8 points worse. Half B fits badly: χ^2/dof is 149–211 at 62–74 MHz, against 52–67 in half A. Most of that comes from the antenna parked looking down, 07-16 23:30 to 07-17 01:05 UTC, which also holds the solar burst of

§6.7. Without the park (2248 bins), half B fits at χ^2/dof 14–24, but reflection then gains at most 0.02 points over a constant ground at 74–82 MHz. Splitting half A by day at 07-16 00:00 UTC gives 07-15 (830 bins) and 07-16 (2559 bins), with comparable spread in ground fraction. R is 1.34–1.52 on 07-15 and −0.06 to 0.17 on 07-16 (errors up to 0.06). On 07-16 a shifted sky is the best model at 6 of 6 frequencies.

Window selection. `global_fit.py` drops gain windows whose median residual over the fitted frequencies is above 5% of the power, so which windows enter depends on the frequencies fitted. Gated on 62–82 MHz alone, 6066 bins in 44 windows pass instead of 6816 in 46. Half B then loses the look-down park (2677 of its 3405 bins remain) and fits at χ^2/dof 21–79. Over all of phase C, reflection then gains 0.17–0.25 points over a constant ground, and the shifted skies at most 0.02. The free-level 6 h sky no longer beats the tied real sky at 62–70 MHz (−0.27 to −0.08 points). The day split is the same under either gate.

Faster gain. The fits above put a gain knot every 3 h, and §7.6 finds the gain changing within the hour. `validate_phase_c_fastgain.py` refits phase C with knots every 3 h, 1 h and 20 min, and injects that sub-hour gain into mocks (Table 12; data-analysis branch `daily-term-fast-gain-v0004`, commit ad209ab, pushed and not merged). Every fit anchors the GSM amplitude in tcal kelvin at $a_{\rm tcal} = 1.5$, this model's level at 62–82 MHz with 3-h knots, either softly (±0.02 in log, which these data outweigh) or hard ($\pm 10^{-4}$).

- **With faster knots the term is no longer needed.** With 3-h knots, reflection with the real sky beats a constant ground by 0.24 to 0.30 points. With 1-h knots the difference is −0.02 to 0.03, and with 20-min knots −0.03 to 0.04. Freed this way, the gain level falls to $a_{\rm tcal} = 0.65$–0.84 against the soft anchor. Held at 1.5 by the hard anchor, reflection is fitted with $R = 0.46$–0.76 at 1-h knots but does no better than a constant ground (−0.19 to 0.02 points).
- **The faster gain takes up the sky's daily shape.** Per gain segment, the log gain knots of the constant-ground fits correlate with $\log \bar{S}$ of the real sky at +0.37 to +0.75 (1-h knots) and +0.43 to +0.85 (20-min knots), and with the shifted skies at −0.37 to +0.18 and −0.53 to +0.08. With 3-h knots the real sky gives +0.00 to +0.27.
- **The sub-hour gain does not make the term.** The mocks take the 3-h fits as truth, times the frequency-flat fast gain of §7.6 (3.2% rms, its temperature-driven part included). Without a daily term, reflection gains 0.01 to 0.05 points with 3-h knots. With the real data's own residuals multiplied in it gains −0.03 to 0.02, against the real data's 0.24–0.30.
- **Faster knots would hide a real term.** A mock with a daily term ($R = 0.60$–0.84), the fast gain and the real residuals keeps its advantage with 3-h knots (0.13 to 0.26 points). With 1-h knots it drops to −0.29 to 0.08 points, although R is still recovered (0.58–0.76). Without the residuals it keeps a clear advantage with 3-h and 20-min knots (0.35–0.48 and 0.38–0.62 points), and a weak one with 1-h knots (−0.00–0.16). With 20-min knots and the residuals, the comparison is erratic in both mocks: −0.10 to 0.50 points without a term and −0.12 to 0.53 with one. So the real data's loss of the term with faster knots is what a real term would also show, given the residuals these fits leave.

	3 h knots		1 h knots		20 min knots	
	gain	R	gain	R	gain	R
Real data, soft anchor	+0.24 to +0.30	0.60 to 0.84	−0.02 to +0.03	0.01 to 1.00	−0.03 to +0.04	−0.28 to 0.86
Real data, hard anchor	–	–	−0.19 to +0.02	0.46 to 0.76	−0.34 to −0.05	0.28 to 0.77
Mock, no daily term, fast gain	+0.01 to +0.05	0.09 to 0.26	−0.10 to +0.00	0.08 to 0.48	−0.23 to −0.03	0.07 to 0.46
Mock, daily term, fast gain	+0.35 to +0.48	0.60 to 0.80	−0.00 to +0.16	0.60 to 0.80	+0.38 to +0.62	0.59 to 0.83
Mock, no daily term, fast gain and residuals	−0.03 to +0.02	0.04 to 0.32	−0.14 to +0.04	−0.13 to 0.61	−0.10 to +0.50	−0.22 to 0.71
Mock, daily term, fast gain and residuals	+0.13 to +0.26	0.61 to 0.86	−0.29 to +0.08	0.58 to 0.76	−0.12 to +0.53	0.42 to 0.96
Gain knots against $\bar{S}$: real sky	+0.00 to +0.27		+0.37 to +0.75		+0.43 to +0.85	
shifted skies	−0.43 to +0.45		−0.37 to +0.18		−0.53 to +0.08	

Table 12: The daily term against faster gain (`validate_phase_c_fastgain.json`), at 62–82 MHz. "gain": held-out rms of a constant ground minus that of reflection with the real sky, in percentage points (positive: the daily term helps), over the converged fits; R: the fitted reflection coefficient. Soft and hard anchors hold $a_{\rm tcal}$ at 1.5 by ±0.02 and $\pm 10^{-4}$ in log. The mocks are the 3-h fits times the fast gain of §7.6, with radiometer noise, and in the last two rows with the real fits' residuals multiplied in (rolled by 1500 bins). Bottom: the correlation of the constant fits' log gain knots with $\log \bar{S}$, per gain segment.

What this shows. Over phase C, with gain knots every 3 h, rx-A raw power at 62–82 MHz carries a slow additive daily term that a constant ground misses. It does not follow the sky of the current pointing, none of the five housekeeping streams reproduces it, and the measured sub-hour gain does not make it. With gain knots every hour or faster the data no longer need it, because the gain then follows the same daily shape. A mock with a real term behaves the same way, so faster knots do not rule the term out. Whether it is additive at all, or a gain change on 1–3 h scales that the 3-h knots miss, these fits cannot tell. Where the receiver's gain after the RF switch is measured, on 07-17 at 87.5 m, it has no 1–3 h structure that follows the sky (§7.6). That stretch shows no daily term either, even with 3-h knots, so it does not say what carries the term where it is found. A gain change could arise there before the switch, in the antenna, balun or cable, or after it at times without calibration cycles; otherwise the faster knots absorb a mismatch in the sky or beam model. The 30 m era, where the term is found, has one usable calibration interval, 07-16 01:41–02:24 UTC with the antenna moving (§7.6). It lies in half A, on 07-16. Over it the gain after the switch changes by at most 0.9% at 110–205 MHz, against −0.7 to 7.1% for the 20-min fit's gain over the same minutes; at 43 min it is far too short to show a daily term itself. That it has the sky's time shape is not established, and neither is a reflection coefficient. Both the real-sky preference and R change from one day to the next, under either window gate. The sidereal-time controls also stop discriminating once the ground's level is freed, although that depends on which windows enter. Over phase C, sidereal time gains only 3.9 min a day on solar time, so these data cannot separate a sky term from an instrumental one with a daily cycle. That needs observations weeks apart.

MHz	constant	real	+6 h	−6 h	12 h	beam	pickup	R (real)
62.5	6.18	5.93	6.13	6.16	6.19	6.16	**5.86**	0.68±0.03
66.4	6.52	6.25	6.31	6.51	6.52	6.51	**6.13**	0.65±0.03
70.3	6.02	5.71	5.76	6.00	5.99	6.02	**5.69**	0.60±0.03
74.2	5.49	5.20	5.33	5.48	5.43	5.48	**5.13**	0.62±0.04
78.1	5.33	**5.06**	5.35	5.33	5.30	5.32	5.10†	0.70±0.04
82.0	6.12	**5.86**	6.30	6.13	6.13	6.09	6.09†	0.83±0.05

Table 13: The phase-C fits on all bins (`validate_phase_c.json`): held-out rms in percent of the power for each ground form (one free parameter each; "real" and the shifted skies are the reflection form), and the reflection coefficient R of the real sky. Bold: the best fit in the row. †: the fit diverged (a gain knot below 1% of the median, or an offset above ten times the GSM column).

7.6 The ground term with a fast gain

The phase-C fits above hold the gain smooth over 3 h. Freeing it on shorter timescales changes the ground term several-fold, so this subsection asks which gain the data support and what ground term follows. `global_fit.py` (data-analysis branch `sky-terrain-model-v0004`, commit a7f7415, pushed and not merged) fits rx-A in `raw_binned_v0001` over phase C at 21 frequencies from 51 to 234 MHz, one at a time: a smooth gain times [GSM through the HFSS beam + $T_{\rm gnd}$ times the ground fraction + an offset + the transmitter's leakage + the Sun], with sky corrections to $l = 2$. The gain prior allows 10% per knot step. Bins during the 07-16 23:50–00:02 UTC solar flare (§7.5) are excluded, which leaves 6321 bins in 46 windows. Alternate 5 min blocks are held out. `validate_ground_term.py` in this directory runs every fit quoted here.

Which gain. With knots every 20 min instead of 3 h, $T_{\rm gnd}$ at 117–180 MHz rises 2.2–6.7-fold (213 K to 470 K at 148 MHz; at 1 frequency the 3-h value is negative), while at 203 MHz the two agree (224 and 185 K). The held-out rms falls by 0.1–0.9 points. Held-out blocks score how well each model predicts nearby bins, not whether its $T_{\rm gnd}$ is right, so the choice is tested by injection (Fig. 17). A mock is one fit's own prediction for every bin, with radiometer noise, times the 20-min fit's real residuals rolled in time, so they keep their size and timescale but not their alignment with pointing. Its $T_{\rm gnd}$ and gain are then known.

- A mock with the 3-h fit's slow gain is recovered by both knot spacings, the 20-min one to 99 K rms (at most 217 K; 375 K at 148 MHz). Freeing the gain does not by itself inflate $T_{\rm gnd}$.
- A mock with the 20-min fit's gain, refitted with 3-h knots, returns the low values the real data give with 3-h knots, within 109 K at every frequency (188 K at 148 MHz).

The real data behave like the second case: the model needs a multiplicative term that changes within the hour, and slow knots leave those changes to be absorbed by the ground and offset terms. Below, it is called the fast

MHz	constant	real, tied	real, free	+6 h	−6 h	12 h	R (real, tied)
Half A							
62.5	4.24	4.08	4.23	5.06	**4.07**	4.24	0.77±0.04
66.4	3.38	**3.13**	3.25	4.37	3.28	3.35	0.67±0.03
70.3	3.51	**3.33**	3.51	4.94	3.45	3.46	0.56±0.03
74.2	4.10	**3.93**	3.95	5.75	4.07	4.01	0.54±0.03
78.1	4.94	4.66	**4.63**	6.32	4.90	4.85	0.71±0.05
82.0	4.81	**4.45**	4.46	5.51	4.69	4.74	1.25±0.11
Half B							
62.5	10.12	9.37	10.39	**9.18**	10.20	10.14	0.57±0.08
66.4	10.51	9.71	10.86	**9.42**	10.64	10.59	0.60±0.10
70.3	7.63	**7.29**	8.01	8.00	7.88	7.93	0.50±0.12
74.2	5.69	**5.68**	6.58	6.33	5.96	6.27	0.77±0.18
78.1	**3.49**	3.51	3.52	3.54	3.50	3.51	0.35±0.14
82.0	2.19	**2.15**	2.21	2.39†	2.21	2.18	0.17±0.16
07-15							
62.5	5.58†	**5.44**	–	5.42†	5.61†	5.73	1.50±0.12
66.4	5.06†	**4.94**	–	4.87†	5.13†	5.18	1.52±0.12
70.3	4.47†	**4.36**	–	4.29†	4.49†	4.76	1.43±0.08
74.2	4.86	**4.25**	–	4.60†	5.65	5.00	1.34±0.05
78.1	5.76	5.18†	–	5.02†	7.23	**5.55**	1.34±0.07†
82.0	6.68	6.54†	–	6.32†	8.00	**6.56**	1.48±0.13†
07-16							
62.5	3.60	3.50	–	**3.43**	3.57	3.55	0.17±0.06
66.4	3.03	2.99	–	3.04	**2.95**	3.02	0.15±0.04
70.3	3.16	3.16	–	3.21	3.16	**3.13**	0.05±0.04
74.2	3.68	3.71	–	3.73	3.70	**3.57**	0.02±0.05
78.1	4.45	4.51	–	4.48	4.50	**4.31**	-0.06±0.06
82.0	3.69	3.69	–	3.55	**3.53**	3.70	0.10±0.06

Table 14: The same comparison on disjoint parts of phase C (`validate_phase_c.json`). Half A is phase C before 07-16 18:10 UTC, half B after it. 07-15 and 07-16 split half A at midnight UTC. "real, free" and, in the halves, the shifted skies have a free level with the 290±300 K prior; in the two days the shifted skies are tied, like the real sky. Bold: the best converged fit in each row; †: diverged.

gain; where the receiver's own gain is measured it is not that ("The receiver's own gain"). The 20-min fit is the one used below.

The ground term and its error (Table 15). At 110–205 MHz $T_{\rm gnd}$ falls from 693 K to 185 K. The error is the rms over 3 injections of recovered minus true $T_{\rm gnd}$ (two rolls of the residuals on the 20-min mock, one on the 3-h mock): 26–323 K, 0.7–7.6 times the fit's own error (17–141 K), which takes the residuals as white. The draws average 38 K. With three draws per frequency the error is a scale, not a precise σ. The residuals in the mocks do not follow the pointing. Real residuals that do, from terrain or beam errors, could bias $T_{\rm gnd}$ in a way these injections do not test. This ground term and the rotating antenna's (Table 10) both fall with frequency over 125–175 MHz, but they come from different models, beams and data, and are not combined.

Where the fit collapses. At 51, 66, 74, 188, 195, 211, 219, 227 MHz the 20-min fit has two solutions of nearly equal χ^2. In one, $a_{\rm tcal}$ is near its values elsewhere in the band (1.7–16.4). In the other the gain collapses, to $a_{\rm tcal}$ =0.001–0.29 (at 1 of them the fit is flagged diverged instead), and the ground, offset, transmitter and Sun terms grow to keep the level. The fit finds the second. Holding $a_{\rm tcal}$ near 4.5 with a prior of ±0.2 in log brings it back above 0.50 at 5 of these 8 frequencies (0.3–4.1), and changes χ^2/dof by at most 0.27% and the held-out rms by at most 0.04 points. The prior is soft: its rows are not scaled by χ^2/dof, so these data can outweigh it (§7.5 uses a hard one). Where it does pull $a_{\rm tcal}$ back, the fit is no worse, so the data do not choose between the two solutions there. The sky is in those data. In the longest parked stretch (07-16 02:48–16:31 UTC, 30 m, 961 bins), raw power at these frequencies tracks the GSM column with correlations of 0.62–0.97, with no gain model at all; in a shorter 87.5 m stretch (07-17 16:29–18:50, 145 bins, where the sky changes less) 4 of the 8 exceed 0.5. In the long stretch the slope against the column is 5–2261 times the gain the collapsed fit puts there, where at the other frequencies the ratio is 0.20–1.4. In the collapsed solution another term carries the sky's daily drift. Below 90 MHz it is the 20-min gain: its log swings by 32–64% rms over phase C and correlates with the log GSM column at 0.34–0.85, where in the table's band it swings by 9–26%. Above 180 MHz the transmitter and Sun coefficients grow at least 22-fold. Then only the contrast between pointings ties the GSM's amplitude

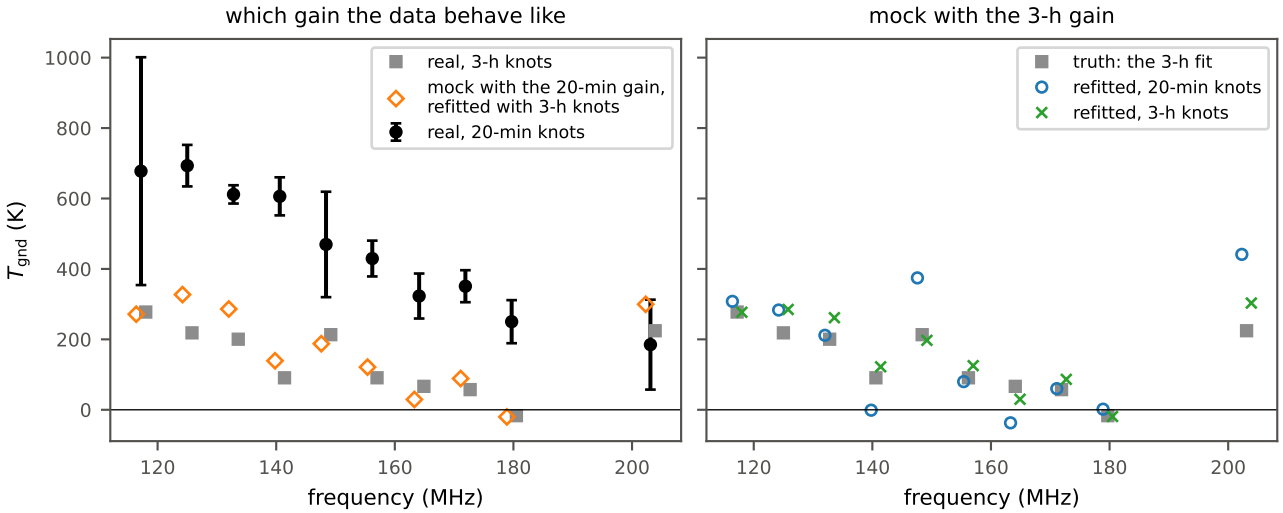


Figure 17: T_{gnd} at 110–205 MHz (rx-A, `raw_binned_v0001`, `validate_ground_term.json`; frequencies where the 20-min fit collapses omitted). Left: the real data with 20-min gain knots (errors from the injections) and 3-h knots, and a mock with the 20-min fit's gain refitted with 3-h knots, which reproduces the real 3-h values. Right: a mock with the 3-h fit's gain (its truth in grey), recovered by either knot spacing. Points are offset by 0.8 MHz for legibility.

MHz	T_{gnd} (K)	injection (K)	fit (K)	a_{tcal}	
50.8	–	–	–	0.286	collapsed
58.6	4032	2320	186	8.75	unstable
66.4	–	–	–	0.013	collapsed
74.2	–	–	–	0.002	collapsed
82.0	1817	507	298	5.22	
117.2	678	323	141	1.73	
125.0	693	59	47	4.69	
132.8	612	26	39	4.63	
140.6	606	54	36	4.44	
148.4	470	150	25	4.62	
156.2	430	51	19	5.42	
164.1	323	64	17	6.06	
171.9	351	45	35	3.66	
179.7	250	61	41	3.11	
187.5	203‡	52	–	2.99‡	shared-gain fit
195.3	198‡	13	–	3.14‡	shared-gain fit
203.1	185	128	17	8.08	
210.9	–	–	–	0.048	collapsed
218.8	–	–	–	0.001	collapsed
226.6	–	–	–	0.981	collapsed
234.4	387	491	144	16.44	unstable

Table 15: T_{gnd} over phase C with 20-min gain knots (rx-A, `validate_ground_term.json`). "injection": the rms of recovered minus true T_{gnd} over the injections, the error to use; "fit": the fit's own error. a_{tcal} is the GSM amplitude, gain times `tcal`'s scale. "collapsed": a_{tcal} below 0.50 (or diverged), where no T_{gnd} is quoted. "unstable": collapses once the transmitter's bins are dropped. ‡: a different model, the fast gain shared within each band ("A fast gain shared across frequencies", `validate_shared_gain.json`), quoted at 188, 195 MHz, where the per-frequency fit collapses. Its error is the injection rms over 3 mocks of that model.

down, and at these frequencies that contrast is spoiled. Below 90 MHz the residual reaches 4.8% in the worst window (median 1.0% over the windows), where 148–180 MHz has 0.6% (median 0.6%). Above 180 MHz the 4 windows with the transmitter on reach 10.6%, against a median of 0.8% in the others. Dropping transmitter bins (5656 bins left) rescues 74, 211 MHz but turns 59, 234 MHz into collapsed fits. χ^2/dof is 7.7–50.0 where the fit collapses, against 4.6–12.1 at the frequencies of the table. These frequencies are left out of the ground term. They include 62–74 MHz, where the daily term of §7.5 was found, so that analysis cannot be repeated with the fast gain. With faster knots there the gain takes up the daily term, as it does the sky here (§7.5, "Faster gain").

What the fast gain follows. Take the 20-min fit's log gain, average it over the frequencies of the table, and subtract a 3-h piecewise-linear fit: what is left varies by 3.2% rms over 138 knots, and the averages of two disjoint halves of those frequencies correlate at 0.90. `gain_drivers.py` projects every housekeeping stream in the correlator metadata onto the same knots, detrends it the same way and compares the two in two disjoint stretches: 07-15/16 at 30 m, and 07-17 at 30, 87.5 and 91 m (Fig. 19). On 07-17 the gain falls as the RF switch warms. The robust (Theil–Sen) slope is −2.8%/K. A bootstrap over blocks of 6 knots, which respects the correlation between neighbouring knots, puts the least-squares slope between −4.0 and −1.3%/K (95%), and the medians fall from the coldest to the warmest fifth of the knots (2.0 to −2.5%). On 07-15/16 the sign is the same (−2.8%/K; −4.2 to −0.6%/K), but the four cooler fifths have median fast gain between −0.9 and 0.9%, with no trend among them. Most of the slope comes from the warmest fifth (−3.2%), so that stretch constrains the relation only weakly. There is no lag at 20-min resolution; the switch's fast temperature varies by 0.6 and 0.5 K rms. The other two switch thermistors and the load's temperature give the same. No other stream (the system current, the IMU, the motors, the pot, calibration and VNA cycles, the flag occupancy) correlates with the same sign at better than $|r| = 0.20$ in both stretches. Around the end of calibration cycles, VNA sweeps and antenna moves, the 3-h fit's residual moves by at most 0.5% (1.3 times its standard error), against up to 0.3% around random times. A temperature term does not replace the fast knots. Adding the switch's temperature as a regressor to 3-h knots, as one term or as a slow and a fast term, leaves T_{gnd} near the 3-h values, and refits a mock that carries the rest of the fast gain far below its truth (`gain_drivers.py` and the data-analysis README record those runs). 53–71% of the fast gain's variance has no recorded cause.

The receiver's own gain. The calibration visits of `tcal@v0003` measure the gain after the RF switch without any sky model: the load's power over its temperature, and the noise source's excess over the load, which also carries the source's own drift. `validate_cal_gain.py` compares them with the fits (data-analysis branch `cal-gain-daily-term-v0004`, commit 8530b04, pushed and not merged). They are dense only on 07-17, 04:11–16:22 UTC at 87.5 m, with the antenna mostly parked: 59 pairs 12 min apart. The 30 m era has too few to test, and nothing before the switch (antenna, balun, cable) is seen. Over that stretch (Fig. 18) the measured gain at 110–205 MHz varies by 0.8–1.3% rms by the load and 2.8–2.9% by the noise source, and follows the switch's thermistor (30.3–33.4 °C): the band's common mode changes by −1.2%/K by the load ($r = -0.92$) and −2.5%/K by the noise source ($r = -0.77$), less steep than the fast gain's. After a running 3-h mean is subtracted, its correlation with the GSM column is −0.20 to 0.11 at every frequency: no 1–3 h structure follows the sky. Its common mode then varies by 0.5% rms (noise source), against 1.6% for the 20-min fit's gain (correlation 0.31). Over the same hours the 20-min fit's gain at the table's frequencies moves by 7.4–27.7% rms and follows the GSM column, with correlations up to 0.99 (117 MHz) and above 0.5 at 3 of the 10 frequencies of the table. In this parked stretch the fast gain is therefore not the receiver's gain after the switch. It is a gain change before the switch, or the fit absorbing a mismatch in the sky or beam model at a fixed pointing. The 30 m era has one usable interval. The raw correlator files of 07-16 (`validate_cal_thirty.py`; 39 files, SHA-256 recorded) hold 3 load visits. A fixed rule drops a visit that lies more than 10% from the visits' median at more than 2 of the 21 frequencies. It drops the first, at 01:14 UTC with only 22 rows, which fails at 6. Those frequencies all lie within 0.125 MHz of an integer MHz, and every frequency where it passes at least 0.19 MHz from one, as contamination by a comb with 1.000 MHz spacing would give. Memo 001 dates the onset of box-air's 1.000 MHz comb on 07-16 to 01:16:57, after this visit, so its source is not identified. Between the other two, 01:41–02:24 UTC (43 min, the antenna moving through the 07-16 raster), the load gain after the switch changes by −0.9 to 0.5% at 110–205 MHz and by at most 0.6% at 51–82 MHz. The 20-min fit's gain changes over the same minutes by −0.7 to 7.1% at 110–205 MHz and 8.2–18.9% at 51–82 MHz, while the GSM column rises 6.1–9.9%. Over that one interval, too, the fast gain is not the gain after the switch. With the antenna moving, it may there absorb a mismatch that follows the pointing rather than the sky, and one interval says nothing about 1–3 h structure or about the nights where the daily term of §7.5 is found. Fixing the gain to either calibration series,

with one free level (Table 16), cannot test the daily term of §7.5 on 07-17. That stretch has none even with free 3-h knots (reflection with the real sky against a constant ground: −0.09 to 0.04 points). With the gain fixed, the held-out residual rises from 0.39–1.38% to 1.21–3.31%, and a shifted sky beats the real one at 6 (load) and 6 (noise source) of the 6 frequencies: the calibration gain is not accurate to the 1% this test needs.

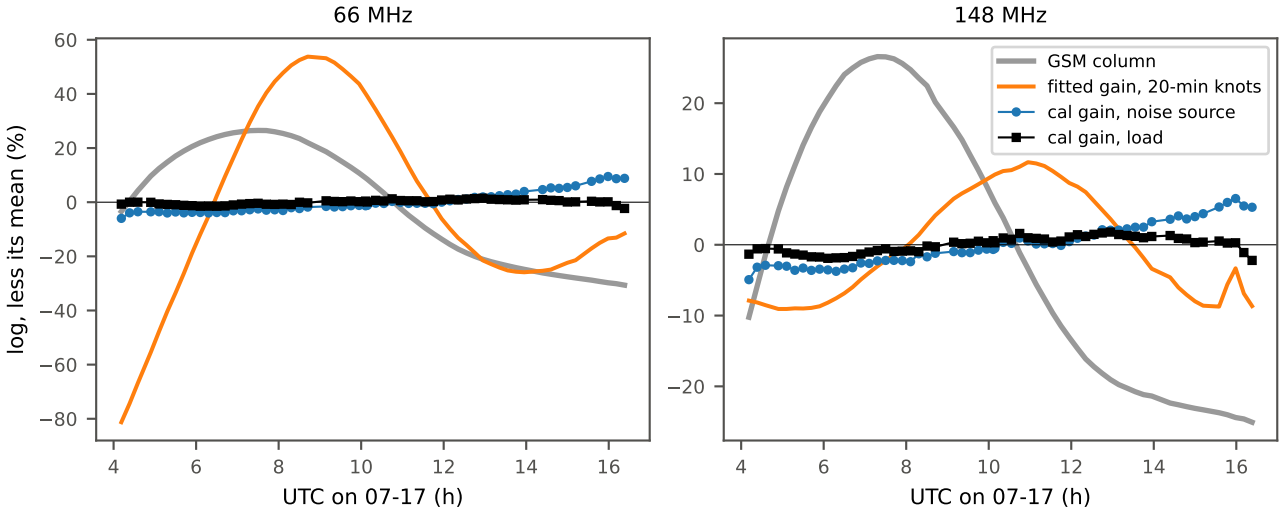


Figure 18: The receiver's gain after the RF switch from the calibration visits (`tcal@v0003`; load power over load temperature, and noise source minus load), the 20-min fit's gain at the same times, and the GSM column through the beam, each as a log less its mean, on 07-17 at 87.5 m (`validate_cal_gain.json`). At both frequencies the fitted gain moves far more than the measured gain; at 66 MHz it follows the sky.

	3-h knots			fixed, noise source			fixed, load		
MHz	const	real	shifted	const	real	shifted	const	real	shifted
62.5	1.01	−0.04	−0.18	3.31	−0.34	+1.67	2.14	+0.45	+1.99
66.4	0.47	−0.09	+0.01	3.13	−0.29	+2.58	1.60	+0.39	+1.36
70.3	0.56	−0.02	−0.01	3.29	−0.33	+2.41	1.25	+0.39	+0.68
74.2	0.49	+0.04	−0.05	3.01	−0.33	+2.13	1.31	+0.38	+0.78
78.1	0.39	−0.01	−0.02	2.58	−0.34	+2.08	1.58	+0.37	+1.35
82.0	0.48	+0.01	−0.02	2.76	−0.30	+2.63	1.21	+0.33	+0.85

Table 16: The daily term on 07-17 04:11–16:22 UTC (87.5 m, 628 bins; the phase-C model of §7.5 with $l = 0$; `validate_cal_gain.json`): held-out rms of a constant ground (%), and the improvement on it of reflection with the real sky and with the best of the ±6 h and 12 h shifted skies (percentage points). Gain free with 3-h knots and a soft amplitude anchor, or fixed to the calibration series from the noise source or the load, with one free level.

A fast gain shared across frequencies. The 20-min fit's fast gain is nearly the same at every frequency ("What the fast gain follows"), so one fast term could serve many frequencies, and a frequency could then no longer collapse by letting its own gain follow its sky. `validate_shared_gain.py` (data-analysis branch `sky-terrain-model-v0004`, commit 3e7b805, pushed and not merged) fits each frequency with a 3-h gain times one fast gain shared across the training frequencies: knots every 20 min, less what the 3-h gain can carry, with a gain and a tilt in frequency, or a quadratic in $\log \nu$. The shared gain and every training frequency's parameters are fitted jointly (Levenberg–Marquardt), until their steps are at most 1.4×10^{-6} in log gain (at most 300 iterations). An earlier version solved them in turn and did not converge, and its band comparison does not survive convergence. One selection rule is needed: a training frequency that collapses during the fit ($a_{\mathrm{tcal}} < 0.50$, or diverged) is dropped from the training set and the fit repeated, since otherwise it wanders along its own flat direction and drags the shared gain. The fits drop, trained on 117–203 MHz: none (linear), 188,

195, 203 MHz (quadratic); trained on 51–203 MHz: 74, 203 MHz (linear), 51, 203 MHz (quadratic) and none (split, below). The injection refits apply the same rule. Of the 12, the ones that drop exactly what the real fit with the same training drops number 0 of 6 with 117–203 MHz training and 0 of 6 with 51–203 MHz. They drop 203 MHz in 7, 188 or 195 MHz in 1 and a low-band frequency in 2. So which frequencies collapse depends partly on structure in the real data that the mocks, built from the model's own prediction, do not carry.

Shared within each band, with its own gain and tilt below and above 100 MHz, the fast gain removes the collapse at 51, 66, 74, 188, 195 MHz (5 of the 8; a_{tcal} =0.9–3.1, marginally at 74 MHz). There its held-out rms is −0.87 to −0.07 points from the per-frequency fit's, and at the frequencies of Table 15 −0.63 to 0.19 points. Shared across both bands with a gain and a tilt, it gives nearly the same high-band ground term as the high band alone. Trained on 51–203 instead of 117–203 MHz, T_{gnd} changes by at most 28% at 117–164 MHz (579 and 695 K at 148 MHz), and by up to a factor 1.7 at 172–195 MHz. A quadratic in $\log \nu$ is less stable. Trained on 117–203 MHz it drops the top of the band and extrapolates there, and the two trainings then differ by up to a factor 7.3. Against mocks built from the two quadratic fits, with three rolls of the real residuals and the same drop rule, that difference exceeds the injection scatter at 6 of 10 frequencies and is within it at 117, 148, 156, 164 MHz. So with a converged fit and a linear frequency dependence, sharing the fast term across both bands barely changes the ground term at 117–164 MHz, and nothing here requires a band-specific fast term. At the frequencies it rescues, mocks of the split fit itself (three rolls of the real residuals, the same drop rule) give its error. A value enters Table 15 only if no refit drops or collapses there and the injection rms is below 50% of the value. That admits 188, 195 MHz, with injection rms of 13.2–51.9 K, marked in the table as coming from this model; frequencies it does not admit: none. The rest of the table is the per-frequency fit, and the two models differ: at 180 MHz the split fit gives 209 K where the table's per-frequency value is 250 K, so the marked rows continue the split fit's spectrum, not the table's.

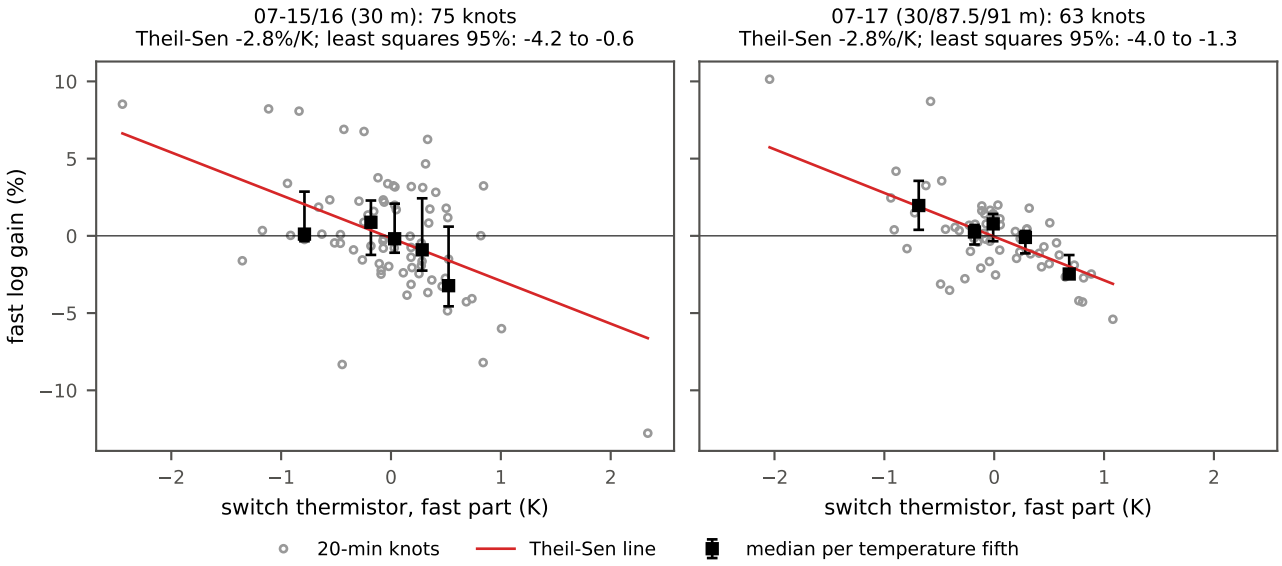


Figure 19: The fast log gain (20-min knots, mean over the frequencies of Table 15, less a 3-h piecewise-linear fit) against the RF switch's thermistor treated the same way, in two disjoint stretches of phase C (`validate_ground_term.json`). Grey: every knot; black: the median and interquartile range in each fifth of the temperature range; red: the Theil–Sen line, with the 95% block-bootstrap range of the least-squares slope above each panel. On 07-15/16 the four cooler fifths show no trend.

7.7 Low-order sky corrections

The same regressions can add corrections to GSM: real spherical harmonics up to $l = 3$ in Galactic coordinates, whose fitted amplitudes over the GSM coefficient are corrections in GSM kelvin. Over an hour the sky turns only about 15° relative to the ground. So the same harmonics frozen to the ground (terrain, horizon or beam errors would look like that) are fitted too, and both are scored on held-out alternate 2-minute blocks (Fig. 20).

The sky version beats the ground version at 14 of 25 frequencies in the raster, and in 3 of the five windows the corrections do not improve the held-out residual at most frequencies. Sky structure is not identified. With the harmonics in, T_{gnd} is no longer constrained either, since the lowest ones mimic the ground fraction. Telling sky from ground needs rotation at many sidereal times.

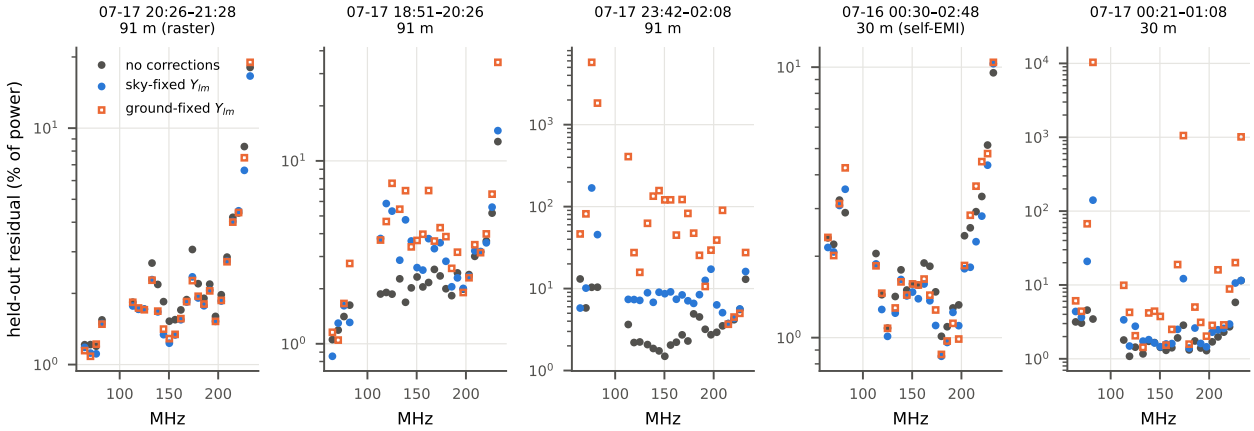


Figure 20: Held-out residual in each moving window, as a percentage of the power, without corrections to GSM (grey), with low-order sky-fixed corrections (blue) and with the same frozen to the ground (orange, open). The corrections do not reliably help: in 3 of the 5 windows (2 of the 3 at 91 m) the uncorrected model has the lowest held-out residual at most frequencies, and the sky-fixed corrections beat the ground-fixed ones at 14 of 25 frequencies in the raster. No sky structure is identified.

8 Open questions

- What is $a \approx 2$: the kelvin scale, the beam, or both? Memo 004's noise-source temperature and memo 005's beam bear on it directly.
- The ground temperature needs calibrated data at more than one height. Extending `tcal` to the 30 m era would give the height lever of Table 1.
- box-gnd stands upright (like box-air at el = 0) with its box on the ground. Its position there is not yet solved. Before it can give a flux scale, a box-air × box-gnd cross-correlation, or a control for the Sun or for heating, its baseline, polarization and differential horizon need substantial work. It is not used here.
- Calibrating the 30 m-era crossings (07-15 to 07-17) would turn the raw-crossing fits of §6.7 into fluxes, and test whether the 07-17 excess is the Sun.
- What makes the ground term rise below 140 MHz? It is not specular reflection, horizon noise or localized sources, and not a fixed fraction of the sky (§7.4). The beam below the horizon, or the antenna's match changing with tilt, are left. A field measurement of S11 or a calibration cycle at several tilts would separate them.
- Separating sky corrections from ground-fixed structure needs rotation at many sidereal times.
- Is the low-frequency daily term over phase C sky or instrumental? Over phase C sidereal and solar time stay nearly in step, and the real-sky preference does not reproduce between days (§7.5). Data weeks apart, or a recorded quantity that tracks it, would decide.
- What is the fast gain? Where the receiver's gain after the switch is measured it is not that, and 53–71% of it follows no recorded stream (§7.6). A gain change before the switch, or the fit absorbing sky or beam errors, is left; calibration cycles in the 30 m era, or a measurement of the antenna path's loss with temperature, would separate them. Until it is known, the gain needs knots every 20 min, and the frequencies where the fit then collapses have no ground term.

- The model is still far from the noise. Candidates are the beam (HFSS against the empirical beam), the sky prior, a time-variable ground, and the 10% GSM prior absorbing structure it cannot predict.

Revision history

2026-10-20 Added a fast gain shared across frequencies (§7.6), fitted jointly to convergence with collapsing training frequencies dropped by rule. It removes the collapse at 51, 66, 74, 188, 195 MHz, and with a linear frequency dependence the high-band ground term changes by at most 28% at 117–164 MHz when the low band shares the gain. The ground-term table adds its values at 188, 195 MHz, marked, with injection errors.

2026-10-19 Moved every fit, product and validation to the extended terrain, DEM v0004 and `horizon_profiles_v0004`: `fit_v0013`–`v0016` and the `_v0004` products replace `fit_v0009`–`v0012` and the `_v0003` ones. No conclusion changes. The phase-C held-out comparisons move by a few hundredths of a point; the largest other moves are the joint-fit floor on T_{gnd} and the night fit's unconstrained T_{gnd} values, both of which the text already treats as unconstrained. Added the 30 m era's one usable calibration interval (§7.6): there too the gain after the RF switch stays within 1% while the fitted fast gain moves several per cent.

2026-10-18 Added the ground term with a fast gain (§7.6): over phase C the gain changes within the hour, injections show 3-h knots bias T_{gnd} low, and with 20-min knots it is quoted at 110–205 MHz with injection errors. The fit collapses at 51, 66, 74, 188, 195, 211, 219, 227 MHz, and the fast gain follows the RF switch's temperature. Where measured (07-17, 87.5 m), the receiver's gain after the switch moves about 1% and does not follow the sky, so the fast gain is not that gain; §7.5 notes what this leaves open for the daily term. Added the low-frequency daily term over phase C (§7.5): a slow additive term at 62–82 MHz that follows neither the pointing nor the recorded housekeeping. Diffuse reflection of the sky, and sky pickup, fit it over the whole phase. The sky origin and the reflection coefficient are withdrawn: the sidereal-time controls work only with the ground's level tied to R, and neither the preference nor R reproduces between 07-15 and 07-16. Faster gain knots (1 h, 20 min) take up the term into the gain, and a mock with a real term behaves the same way, so the term is stated as found with 3-h knots, with its additive nature open. Revised the 07-17 sunset at 30 m. The step in raw power there is mostly a solar burst during a C1.5 flare, ending as the disk goes behind the ridge before the elevation moves. Only the last 18% comes with the elevation step, which the earlier version blamed for all of it.

2026-10-16 Added the effect of extending the DEM 4 km south on the nominal horizon and on the predicted temperature (Section 2, Fig. 1), and the products that follow from it: DEM v0004 (6 km more to the south), `horizon_profiles_v0004` and its coverage check. No fit or product in this memo changed.

2026-10-07 Added the solar bursts, which identify solar emission in box-air; the raw context and raw-power fits of every terrain crossing; the local origin of the daytime impulsive events; the Sun's path through the beam and the raster test. The calibrated 07-17 excess is now read as larger than the Sun, not as the Sun. Added sky and ground recovery: the joint-frequency fit, the sky-calibrated parked nights, and the rotating-antenna regressions with their repeatable but frequency-dependent ground term. Traced the low-frequency excess: not specular reflection, horizon noise or localized sources, not a fixed fraction of the sky; the beam below the horizon or a tilt-dependent match remain, and need a field measurement.

2026-10-06 First version: the model, the degeneracy forecast, the night fit on `tcal@v0003` and `flags@v3`, and the first look for the Sun at the 07-17 terrain sunrise.

A Reproducing this memo

The products are regenerated, from `data-analysis/scripts/marjum-2026-07/ground_sky/` on the data-analysis branch `ground-sky-v0004` (commit 39e130c), with `EIGSEP_CAMPAIGN_ROOT` set to the campaign directory, by the commands below. They read the campaign's pointing table at v2.0, as every earlier version did; the table in the repository has since moved to v3, so check out `marjum-2026-07/curation/pointing_table.*` from commit d2b2205's parent first. Every product that depends on the antenna position reads the horizon profiles `curation/horizon_profiles_v0004` (geometry release `v0004`, memo 002) and traces the Sun on the DEM

mosaic `derived/dem/v0004`, and every product that rotates a beam uses the empirical beam and mount offsets of `empirical_raster_v0013`. The binned spectra, the burst cutouts and the impulsive-event scan depend on neither.

- `python bin_spectra.py v0001` and `python bin_spectra.py v0002 -bin-s 60 -min-rows 10`;
- `python fit_ground_sky.py v0013 -binned v0001 -sun cut-flat`, `v0014 -binned v0001 -sun cut-terrain`, `v0015 -binned v0002 -sun cut-flat` and `v0016 -binned v0001 -sun model`;
- `python sun_events.py $EIGSEP_CAMPAIGN_ROOT/derived/ground_sky/fit_v0015 -out-dir $EIGSEP_CAMPAIGN_ROOT/derived/ground_sky/sun_events_v0004`;
- `python sun_bursts.py v0001`, `python impulsive_scan.py v0002`, `python sun_transitions.py v0004 -sun-events v0004 -fit v0015`, `python sun_profile.py v0004`, and `python raster_sun.py v0004 -beam hfss` then `-beam empirical`. The burst list they read is `curation/solar_bursts_v0001.json`.
- `python joint_fit.py v0004 -fit v0013`; `python sky_cal.py v0005 -beam hfss` and `-beam empirical`, each with `-fit-from 07-15=2026-07-15T05:12:00Z` (data-analysis branch `skycal-0715-cut`, pushed and not merged); `python raster_sky.py v0004` with `-beam hfss -label raster`, `-beam empirical -label raster -reflect`, and for the other moving windows `-beam empirical` with the `-label`, `-start`, `-stop`, `-era` and `-row-step` recorded in each output (07-16 also `-self-emi`); and `python ground_excess.py v0004` with the `-split`, `-label` and window options recorded in each output.

The low-frequency daily term (§7.5) comes from `python validate_phase_c.py -data-analysis <checkout> -work <dir>` in this directory. The checkout is the data-analysis branch `daily-term-fast-gain-v0004`, pushed and not merged, where the code for this negative result stays; `validate_phase_c_fastgain.py` uses the same checkout. That branch is `daily-term-fast-gain` on terrain v0004, which in turn extends `diffuse-ground`. Commits on `diffuse-ground` add the reflection, beam, pickup and housekeeping forms; the remaining options were added at c47da64 (`-ground-group`), fde5185 (`-ground-level-prior`), 6b10518 (`-t-range`), fc1dbb9 (`-night-only`, `-gain-segment-by-era`) and ae990d4 (`-t-exclude`). The 07-17 sunset burst comes from `python validate_sunset_burst.py -data-analysis <checkout>`. Both record their inputs' checksums and the commit they ran at. The ground term with a fast gain (§7.6) comes from `python validate_ground_term.py -data-analysis <checkout> -work <dir>`, with the checkout on the data-analysis branch `sky-terrain-model-v0004` at a7f7415 or later (pushed, not merged). It runs `global_fit.py`, `inject_mock.py`, `hk_streams.py` and `gain_drivers.py` there, each pinned to at most five cores, and records the commit and the binned product's checksum. The receiver's own gain comes from `python validate_cal_gain.py -fit-checkout <dir> -cal-checkout <dir> -fit-work <dir> -cal-work <dir>`, with checkouts of `sky-terrain-model-v0004` and `cal-gain-daily-term-v0004` (both pushed, not merged); it records both commits and the checksums of `raw_binned_v0001` and `tcal@v0003`. The 30 m-era load visits come from `python validate_cal_thirty.py -cal-checkout <dir> -fit-work <dir>`, run after `validate_ground_term.py` with the same work directory, on a checkout of `cal-gain-daily-term-v0004`; it reads raw correlator files and records their SHA-256. The shared fast gain comes from `python validate_shared_gain.py -data-analysis <checkout> -fit-work <dir> -work <dir>` on `sky-terrain-model-v0004` at 3e7b805 or later, run after `validate_ground_term.py` (whose per-frequency residuals its mocks use).

The commits they ran at are below; each product's `manifest.json` records them along with its inputs' checksums. This memo's figures, tables and numbers come from `python make_products.py` in this directory, then `pdflatex memo-008.tex` twice. The script checks the SHA-256 of every product file it reads.

input	SHA-256 (first 16)
`derived/ground_sky/binned_v0001/binned.npz`	`6dc2f340d949dac4`
`derived/ground_sky/binned_v0001/manifest.json`	`259cddcae303f3ec`
`derived/ground_sky/binned_v0002/binned.npz`	`e451dd73d386c739`
`derived/ground_sky/binned_v0002/manifest.json`	`bc244df8e9d42290`
`derived/ground_sky/fit_v0013/fit.npz`	`c65db56cf0fdd3e1`
`derived/ground_sky/fit_v0013/manifest.json`	`5355282132658b48`
`derived/ground_sky/fit_v0014/fit.npz`	`bb125d5c8d30ab83`

derived/ground_sky/fit_v0014/manifest.json	5e1fee68e257de30
derived/ground_sky/fit_v0015/fit.npz	60b9da212882d965
derived/ground_sky/fit_v0015/manifest.json	3d79427a2bbe6288
derived/ground_sky/fit_v0016/fit.npz	38dfc7071d3d95f4
derived/ground_sky/fit_v0016/manifest.json	16afde71345ca532
derived/ground_sky/sun_events_v0004/sun_events.json	86f5b6e52b4a4786
derived/ground_sky/sun_events_v0004/sun_events.npz	41a66688b6a9116b
derived/ground_sky/bursts_v0001/bursts.json	5e69db566e4dc2c0
derived/ground_sky/bursts_v0001/cutouts.npz	bca9c39857fd690e
derived/ground_sky/bursts_v0001/manifest.json	b9f52b6337f844e9
derived/ground_sky/impulsive_v0002/manifest.json	e9b74e567690d724
derived/ground_sky/impulsive_v0002/events.jsonl	a791bb7a24eacc06
derived/ground_sky/transitions_v0004/transitions.json	887388108d01c1e4
derived/ground_sky/transitions_v0004/transitions.npz	062dc0c3f4f4b1d8
derived/ground_sky/transitions_v0004/manifest.json	91cee74e33cff739
derived/ground_sky/sun_profile_v0004/sun_profile.npz	2c776e5b50dca55e
derived/ground_sky/sun_profile_v0004/manifest.json	63e846cf814a7382
derived/ground_sky/raster_sun_v0004/raster_hfss.json	816dd85a9d1b0ee2
derived/ground_sky/raster_sun_v0004/raster_empirical.json	b14b67824d1f59da
derived/ground_sky/raster_sun_v0004/raster_empirical.npz	721eb8aa75591fec
derived/ground_sky/joint_v0004/joint_hfss.json	df30e22557900c49
derived/ground_sky/skycal_v0005/skycal_hfss.json	001a7a6328493024
derived/ground_sky/skycal_v0005/skycal_empirical.json	768596207d1cca9a
derived/ground_sky/skycal_v0005/skycal_hfss.npz	ac24ac173c4e2120
derived/ground_sky/raster_sky_v0004/raster_hfss.json	889751a81ee01148
derived/ground_sky/ground_excess_v0004/raster_horizon_empirical.json	b5ef7a20f0322036
derived/ground_sky/ground_excess_v0004/raster_horizon_noflags_empirical.json	f69d007f74b72f89
derived/ground_sky/ground_excess_v0004/raster_azimuth_empirical.json	29350ac8ca6a0d44
derived/ground_sky/ground_excess_v0004/w0718night91_horizon_empirical.json	9d644083093eaf59
derived/ground_sky/ground_excess_v0004/w0718night91_azimuth_empirical.json	d57bb6e38f9b5928
derived/ground_sky/ground_excess_v0004/w0717eve91_azimuth_empirical.json	24db903013d33f06
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derived/ground_sky/raster_sky_v0004/w0716night_empirical.json	dfba1bc3de2c8890
derived/ground_sky/raster_sky_v0004/w0717dusk30_empirical.json	87c9fd968e864137
marjum-2026-07/curation/horizon_profiles_v0004.npz	1185ddbd39ff5a5f
data-analysis/scripts/marjum-2026-07/ground_sky/summary.json	86d8bff6f2da1566
memos/memo-008-marjum-2026-07-image-domain/inputs/solar_reference.json	e09895904559a8d0
validate_phase_c.json (this memo)	2a6fd5a76fe686cc
validate_sunset_burst.json (this memo)	550a17b9c922a54e
validate_sunset_burst.npz (this memo)	c198e636a7ec7140
validate_ground_term.json (this memo)	623c25c97911f0a0
validate_cal_gain.json (this memo)	17c63f4f85032ba2
validate_cal_thirty.json (this memo)	713862ae7962efe4
validate_shared_gain.json (this memo)	13ee024cb94bb3f0
validate_phase_c_fastgain.json (this memo)	e3a2eafd95718d54
products made with data-analysis	347cecc, 39e130c, 602d7dc, 876c8f9, b3a1b0c, f51af5e
products made with eigsep_base	840c10b
products made with eigsep_data	2b201d4, e941c51
products made with eigsep_sim	d050c83
phase-C fits: data-analysis daily-term-fast-gain-v0004	ad209ab

fast-gain fits: data-analysis `sky-terrain-model-v0004`	`a7f7415`
shared fast gain: data-analysis `sky-terrain-model-v0004`	`3e7b805`
30 m receiver gain: data-analysis `cal-gain-daily-term-v0004`	`0e515b2`
receiver-gain fits: data-analysis `cal-gain-daily-term-v0004`	`8530b04`
eigsep (memo) commit	`278e5ae`
data-analysis commit (degeneracy study file)	`ff35525`
generated	2026-10-21 16:44 UTC